

ROBOCHRONO: A REAL ROBOT BENCHMARK FOR STREAMING TASK UNDERSTANDING

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ABSTRACT

Understanding ongoing robot manipulation requires models to interpret visual observations in relation to interaction history and task progress. We introduce RoboChrono, a benchmark for streaming task understanding comprising 39 scenarios and 34,713 evaluation instances, constructed from real robot executions and complementary bare-hand human recordings. The benchmark evaluates seven tasks grouped into recognition, alignment, and temporal grounding, covering action understanding and anticipation, visual correspondence, temporal ordering, and action localization. Zero-shot evaluation of 18 vision-language models reveals substantial differences across tasks. GPT-6-Astra achieves 98.3% accuracy on Frame Matching but 68.3% on Frame Ordering, while RynnBrain1.1-122B-A10B exhibits a larger gap, reaching 95.4% and 32.9%, respectively. Input ablations on matched questions with five open-weight models further reveal distinct dependencies on visual evidence: removing visual observations reduces Current Action Recognition accuracy by 22.1 percentage points, whereas Next Action Prediction decreases by only 0.7 points. These findings show that strong visual matching does not consistently coincide with strong temporal ordering, and suggest that next-action prediction can be supported by task and action priors even when visual evidence is unavailable. RoboChrono provides a diagnostic setting for examining these differences, highlighting the need for capability-specific evaluation beyond aggregate scores when assessing task understanding in robot manipulation.

1 INTRODUCTION

Understanding an ongoing robot manipulation task requires more than recognizing what is visible in the current frame. An embodied agent must reason over the preceding interaction history to infer the current task state, distinguish completed from ongoing actions, and anticipate what may happen next. Prior work on egocentric video has similarly shown the importance of temporal context for fine-grained action understanding and anticipation (Damen et al., 2018; Furnari & Farinella, 2019; Jia et al., 2022). The problem is particularly challenging from robot-mounted viewpoints, where occlusion, motion, and partial observability can limit the evidence available from any single observation or camera (Jangir et al., 2022; Khazatsky et al., 2024). We study streaming temporal task understanding: interpreting the progress of robot manipulation using only visual observations available up to the queried moment, without access to any subsequent frames or evidence.

Existing benchmarks have advanced several complementary aspects of embodied visual reasoning. Human egocentric benchmarks evaluate temporal and causal reasoning over recorded activities (Jia et al., 2022), while video based spatial benchmarks focus on understanding spatial relationships from visual observations (Yang et al., 2025). Robot centric datasets and benchmarks further connect visual understanding with long horizon reasoning, planning, and physical interaction (Sermanet et al., 2024; Chen et al., 2026a; Bao et al., 2026), and recent embodied foundation model benchmarks broaden evaluation toward multimodal reasoning in physical environments (Ji et al., 2025; Dang et al., 2026). However, these settings largely emphasize reasoning over completed or retrospectively available observations, where later frames may provide evidence for interpreting earlier events. In an ongoing robot execution, such future information is unavailable, and models must infer the current task state, distinguish completed from ongoing actions, recover temporal structure, and anticipate subsequent behavior from the observation history available at the queried moment. This creates a distinct evaluation challenge that is not fully captured by conventional video question answering

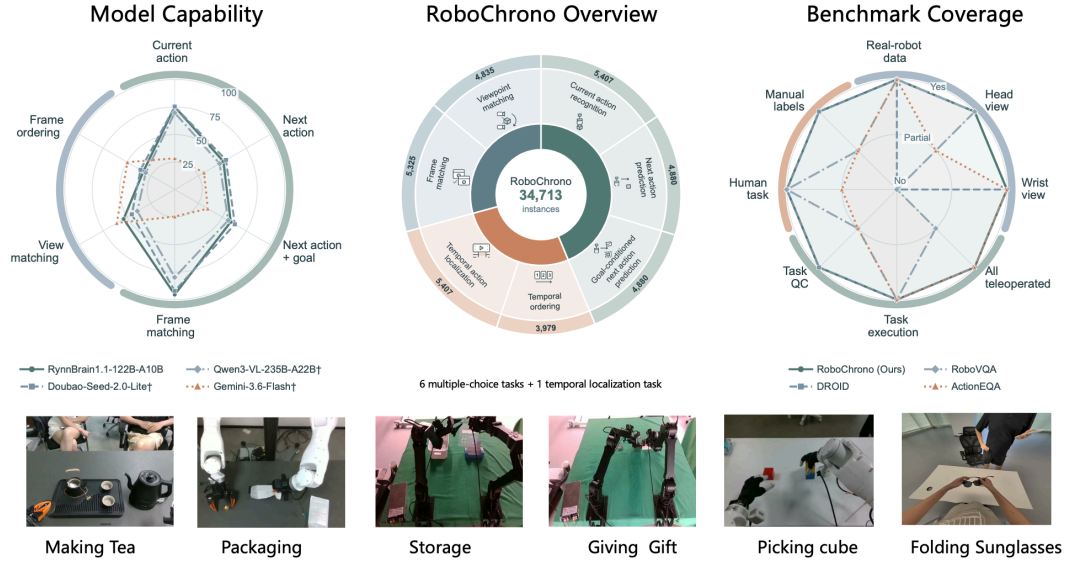


Figure 1: Overview of RoboChrono. Left: Performance profiles of four representative multimodal models across the six multiple-choice evaluation tasks. Center: Distribution of 34,713 evaluation instances across seven tasks. Right: Comparison of data collection and annotation attributes across representative robot benchmarks and datasets.

or full video reasoning. We therefore focus on streaming temporal task understanding and ask whether multimodal models can reliably interpret the temporal progress of real robot manipulation under explicitly restricted observation histories. Table 1 compares RoboChrono with representative egocentric and robot-centric datasets and benchmarks across data sources and annotation types.

We introduce **RoboChrono**, a real-world robot benchmark for streaming temporal task understanding. RoboChrono contains 39 manipulation scenarios, 1,816 successful execution videos, approximately 25 hours of real robot interaction data, and 34,713 evaluation instances. As illustrated in Figure 1, the benchmark evaluates seven tasks covering action interpretation and anticipation, visual evidence association, and temporal organization. In the streaming setting, models can only access observations available up to the queried moment and cannot use future visual evidence. We evaluate 18 open-source and commercial multimodal models across all 39 scenarios under a zero-shot protocol. Our results reveal substantial differences in capability profiles, particularly on temporal ordering, localization, and future action reasoning, which aggregate scores often obscure.

Our contributions are threefold. First, we introduce **RoboChrono**, a large-scale real robot benchmark specifically designed for streaming temporal task understanding. Second, we establish a unified evaluation framework with seven complementary tasks that assess action understanding, anticipation, cross-view and temporal association, ordering, and temporal localization under controlled observation histories. Third, we provide a systematic zero-shot evaluation of 18 multimodal models across all 39 scenarios, revealing persistent limitations in understanding the temporal structure and progression of real-world robot manipulation, particularly in ordering and localization.

2 RELATED WORK

Egocentric Video Understanding. Egocentric video has become an important setting for studying temporal understanding from the first person perspective. Large scale datasets such as Ego4D (Grauman et al., 2022) and Ego Exo4D (Grauman et al., 2024) support research on perception, episodic memory, action recognition, and forecasting. Recent benchmarks further emphasize higher level task reasoning. EgoTaskQA (Jia et al., 2022) evaluates temporal and causal reasoning over actions and state changes, while EgoSchema (Mangalam et al., 2023), EgoPlan Bench (Chen et al., 2026b), and VidEgoThink (Cheng et al., 2024) study long-video question answering, task planning,

Table 1: Comparison of egocentric and embodied robot task benchmarks and datasets. We compare data sources, robot-mounted views, task collection and annotation protocols, and evaluation scale. RoboChrono uses real-robot executions with complementary onboard views and human-defined temporal annotations, evaluated under restricted observation histories.

Benchmark / Dataset	Primary focus	Data source	Robot-mounted view		Task collection			Curation		Evaluation scale
		Real robot	Head	Wrist	All teleop	Task exec.	Task QC	Human task	Manual labels	
(i) Human-worn or scene-centric embodied cognition benchmarks										
EgoTaskQA Jia et al. (2022)	Human task reasoning	✓	✓	✓	✓	✓	△	✓	△	2k videos / 40k QA
VSI-Bench Yang et al. (2025)	Spatial cognition	✓	✓	✓	✓	✓	✓	✓	△	288 videos / 5k+ QA
RoboSpatial Song et al. (2025)	Spatial / affordance QA	✓	✓	✓	✓	✓	✓	✓	✓	1M images / 3M QA
ERQA Team et al. (2025)	Embodied reasoning QA	△	△	△	✓	✓	✓	✓	✓	400 QA
MMSI-Bench Yang et al. (2026)	Multi-image spatial QA	△	△	△	✓	✓	✓	✓	✓	1,000 QA
RynnBrain-Bench Dang et al. (2026)	Cognition + localization	✓	✓	✓	✓	✓	✓	△	△	3,616 clips / 12k QA
(ii) Physical-robot embodied benchmarks and demonstration resources										
JRDB Martin-Martin et al. (2021)	Perception / tracking	✓	✓	✓	✓	✓	✓	✓	✓	64 min / 3.5k tracks
RoboVQA Sermanet et al. (2024)	QA + long-horizon planning	✓	✓	✓	✓	✓	✓	✓	✓	238 h / 829.5k pairs
DROID ³ Khazatsky et al. (2024)	Robot demonstrations	✓	✓	✓	✓	✓	✓	✓	✓	76k traj. / 350 h / 86 tasks
Open-X VQA / Robo2VLM-1 Chen et al. (2026a)	Robot interaction QA	✓	△	△	✓	✓	✓	✓	✓	176k traj. / 684.7k QA
ShareRobot (Aff./Traj.) Ji et al. (2025)	Planning / affordance / trajectory	✓	△	△	✓	✓	✓	✓	△	51.4k inst. / 1.028M QA
ActionEQA Bao et al. (2026)	Action-state reasoning	✓	△	✓	✓	✓	✓	✓	△	2,629 eps. / 8,795 QA
RoboChrono (Ours)	Robot-ego temporal understanding	✓	✓	✓	✓	✓	✓	✓	✓	39 scenarios / 34,713 QA

Definitions. ✓: yes; ✗: no; △: partial or mixed. “Real robot” denotes physical robot data; “Head” and “Wrist” denote global and end-effector-mounted views, respectively. “All teleop” and “Manual labels” indicate fully human-teleoperated executions and human-produced annotations, respectively. “Task QC” denotes execution quality control.

and embodied reasoning. However, these benchmarks are primarily built from human-worn videos, whose viewpoints and motion patterns differ from those of cameras mounted on physical robots.

Robot-Centric Embodied Understanding. Recent work has extended visual reasoning from human activities to robot interaction data. RoboVQA (Sermanet et al., 2024) constructs large-scale video question answering data from both human and robot embodiments to study long-horizon robotic reasoning. Robo2VLM (Chen et al., 2026a) converts real robot trajectories into visual question answering samples by extracting manipulation phases and supervision from robot states, end-effector poses, gripper measurements, and force signals. ActionEQA (Bao et al., 2026) evaluates the relationship between semantic actions and physical state transitions. More recent embodied benchmarks, including RoboBench (Luo et al., 2025), RoboBrain (Ji et al., 2025), and RynnBrain (Dang et al., 2026), further broaden evaluation toward embodied perception, spatial reasoning, affordance understanding, and planning. These efforts demonstrate the importance of evaluating multimodal models on physically grounded robot data, although their data sources, embodiments, camera configurations, and annotation pipelines are often heterogeneous or constructed for broad embodied capabilities rather than controlled temporal task understanding in streaming settings.

Streaming Video Understanding. Most video understanding benchmarks assume that the complete video is available before answering a query. This offline setting differs from embodied agents that must interpret observations while an interaction is still unfolding. StreamingBench (Lin et al., 2026) evaluates multimodal models with questions presented at different timestamps, targeting real-time perception and accumulated context. OVBench (Huang et al., 2024) further organizes on-line video understanding around perception, memory, and prediction over past, present, and future information. Recent work also reveals a tradeoff between immediate perception and long-range memory (Shen et al., 2026). However, existing streaming benchmarks mainly use general videos and human-centered activities rather than physical robot manipulation with explicit task structure.

Different from existing egocentric, robotic, and streaming benchmarks, RoboChrono focuses on streaming task understanding from controlled real robot executions. All episodes are collected under a unified teleoperation and quality control protocol with complementary head and wrist observations. The benchmark covers both gripper and dexterous-hand manipulation and provides human-annotated temporal events for every execution. These annotations support evaluation of task-stage understanding, event localization, question answering, and next-action prediction at meaningful moments during execution. To our knowledge, RoboChrono is the first benchmark to jointly evaluate these properties in a unified real-robot setting under restricted observation histories.



Figure 2: Overview of the task coverage in RoboChrono. The benchmark includes diverse manipulation objects and action primitives drawn from 39 real-world manipulation scenarios.

3 BENCHMARK DESIGN AND EVALUATION

3.1 BENCHMARK OVERVIEW

RoboChrono evaluates whether multimodal models can understand the progress of real-world robot manipulation from progressively revealed visual observations. The benchmark focuses on three capabilities: understanding ongoing actions and anticipating subsequent behavior, establishing correspondences across observations and camera views, and localizing task-relevant events in time. At an evaluation time t , a model receives the observed video prefix $\mathcal{O}_{\leq t}$ and a task-specific query q_t . It must answer using the available observations without accessing future frames. Depending on the evaluation task, the required output is either a multiple-choice answer or a temporal interval. Explicit task goals are provided for goal-conditioned next-action prediction. The benchmark comprises seven evaluation tasks constructed from temporally annotated manipulation executions. Figure 2 illustrates the range of manipulation objects and action primitives covered by RoboChrono, while Figure 3 summarizes the complete benchmark construction and evaluation pipeline, from frame-accurate temporal annotation to VQA generation and unified zero-shot model evaluation.

3.2 DATA COLLECTION AND TEMPORAL ANNOTATION

We design 39 manipulation tasks, including 29 gripper based tasks, 5 dexterous hand tasks, and 5 ego hand tasks. The task set spans object grasping, picking, placing, transferring, stacking, packaging, cleaning, and tool use, covering both object level rearrangement, fine grained hand object interaction. The objects include everyday items, containers, tools, and packaged goods.

Robot task executions are collected through human teleoperation under a shared collection protocol, except for the five barehand scenarios, which are recorded from a person performing the task bare-handed. The dataset, as part of this benchmark, contains 1,816 successful execution videos and approximately 25 hours of real manipulation data. Repeated executions introduce variation in object configurations, motion trajectories, and interaction patterns within each task in a consistent manner.

Visual observations are recorded from robot mounted or near egocentric cameras. Multi view recordings provide complementary head and wrist views, capturing workspace context and local manipulation details, respectively. Single view recordings are also retained. When multiple camera streams are available, they are synchronized within each execution. All recorded episodes are manually inspected after collection before inclusion in the benchmark and associated with task level metadata.

Each execution is annotated with task-relevant action and state intervals. We represent the annotations as $\mathcal{A} = \{(s_i, e_i, z_i)\}_{i=1}^K$, where s_i and e_i denote the start and end times of an interval, and z_i describes the action or task state. Examples include approaching an object, establishing a grasp, moving an object toward a target, completing placement, and reaching a task-specific goal state. We use temporal intervals rather than timestamps to preserve the duration of actions and persistent task states. These annotations provide the reference semantics and temporal boundaries used to sample evaluation points and construct task-specific questions and answers. The benchmark contains 34,713 evaluation instances spanning seven evaluation tasks across 39 manipulation scenarios.

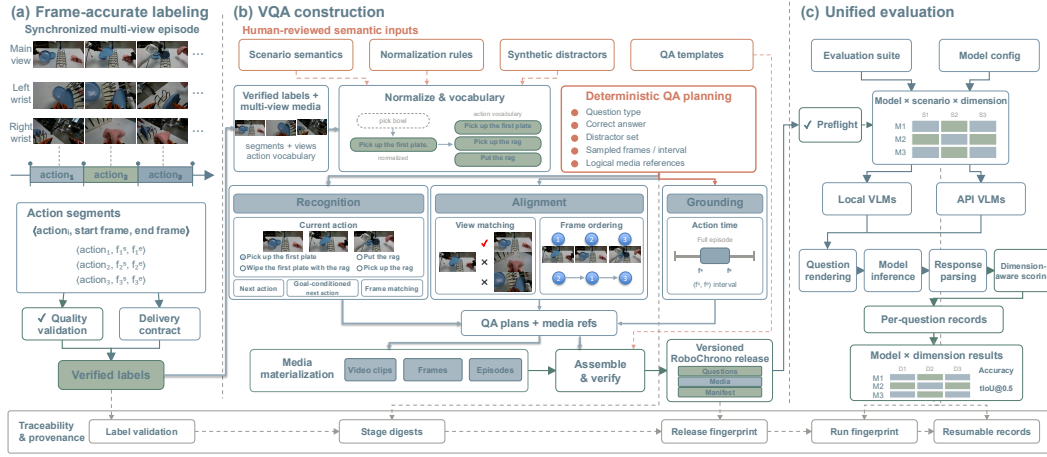


Figure 3: Overview of the RoboChrono construction and evaluation pipeline.

3.3 EVALUATION PROTOCOL

We evaluate all models under a causal observation setting that reflects the information available to an embodied agent during online interaction. For an evaluation instance at time t , the model can only access visual observations available at or before t . Future observations are never included in the model input. For next action prediction, later annotations may be used during benchmark construction to determine the reference answer, but the corresponding future visual evidence remains hidden at inference time. Matching, ordering, and temporal localization instances are similarly constructed using only observations available within the permitted observation range.

The seven tasks instantiate this causal protocol with different forms of visual evidence and reference targets. Table 2 summarizes the model inputs and target outputs for each task.

Table 2: Task inputs and reference targets in RoboChrono.

Task	Model input	Reference target
Current action	Video prefix $\mathcal{O}_{\leq t}$ and answer choices	Action or state at time t
Next action	Video prefix $\mathcal{O}_{\leq t}$ and answer choices	Next action executed after t
Goal-conditioned next action	Video prefix $\mathcal{O}_{\leq t}$, task goal, and answer choices	Next executed action under the goal
Frame matching	Query state or frame and candidate frames from $\mathcal{O}_{\leq t}$	Frame matching the queried state
View matching	Query image and synchronized candidates from other views	Image from another view at the same time step
Frame ordering	Candidate frames from $\mathcal{O}_{\leq t}$	Correct temporal order
Action time localization	Video observation, queried action/state, and video duration	Interval $[s_i, e_i]$ in seconds

View matching is generated only when synchronized multi-view recordings are available; single-view recordings are excluded from this task rather than counted as incorrect. When an action label appears multiple times in one execution, the queried instance is disambiguated by its temporal position and annotation index, so that each question refers to a unique annotated interval.

All models are evaluated in a strictly zero-shot setting without benchmark-specific supervised fine-tuning or adaptation. We use the same task definitions, answer spaces, and scoring rules across models. Prompt templates are kept consistent whenever permitted by the corresponding inference interfaces. Closed-source models are evaluated through their officially available interfaces, while open-source models are evaluated locally using their publicly released checkpoints.

For the six multiple-choice tasks, performance is measured using accuracy. Given N valid evaluation instances, the accuracy is computed as

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i^*], \quad (1)$$

where $\hat{y}_i$ is the model prediction and y_i^* is the annotated reference answer. Task–recording combinations for which a question is not applicable, such as view matching for single-view recordings, are excluded from the corresponding evaluation set rather than treated as incorrect predictions.

For action time localization, the model predicts a temporal interval $\hat{I} = [\hat{s}, \hat{e}]$ corresponding to the queried action or task state. The localization prompt specifies the total duration of the video and requires the predicted interval to be expressed in seconds. We measure the overlap between the predicted interval $\hat{I}$ and the annotated reference interval I^* using temporal Intersection over Union:

$$\text{tIoU}(\hat{I}, I^*) = \frac{|\hat{I} \cap I^*|}{|\hat{I} \cup I^*|}, \quad (2)$$

where $|\cdot|$ denotes temporal duration. Since each query requires a single predicted interval, we report Recall@1 at a tIoU threshold of $\tau = 0.5$:

$$R@1(\tau) = \frac{1}{N_{\text{loc}}} \sum_{i=1}^{N_{\text{loc}}} \mathbf{1}[\text{tIoU}(\hat{I}_i, I_i^*) \geq \tau], \quad (3)$$

where N_{loc} denotes the number of valid localization queries. We refer to this metric as tIoU@0.5 throughout the paper. Finally, we distinguish invalid evaluation runs from valid but low-scoring predictions. A model–task run with a response parse failure rate of 50% or higher is treated as an unsuccessful run and is excluded from reported comparisons. Valid runs are retained regardless of their resulting score, so that poor task performance is not conflated with response parsing failure.

4 EXPERIMENTS AND RESULTS

4.1 EXPERIMENTAL SETUP

We evaluate 18 multimodal models on RoboChrono under the zero-shot protocol described in Section 3.3. All models use the same task definitions and scoring rules, without benchmark-specific supervised fine-tuning. We report accuracy for the six multiple-choice tasks and tIoU for Action Time Localization. The multiple-choice aggregate is computed over the applicable questions and does not include Action Time Localization, which is scored separately by tIoU.

4.2 MAIN RESULTS

Table 3 provides the complete benchmark results. GPT-6-Astra achieves the strongest overall performance by a large margin, reaching 80.5% Choice Avg., compared with 61.2% for RynnBrain1.1-122B-A10B, the strongest open-weight model. Astra leads all seven evaluation dimensions, indicating strong and relatively balanced performance across recognition, prediction, alignment, and temporal grounding. Its strongest result is Frame Matching at 98.3%, showing that it can reliably identify the observation corresponding to the queried visual context. It also performs strongly on Current Action Recognition (89.4%) and Goal-Conditioned Next Action Prediction (83.6%), suggesting effective use of both visual evidence and explicit task goals, though Frame Ordering remains weaker.

However, Astra does not saturate the benchmark. Its performance drops to 76.4% on Next Action Prediction, 68.3% on Frame Ordering, 67.5% on View Matching, and 72.1% on Action Time Localization. These dimensions require reasoning beyond identifying visible content: the model must infer what happens next, recover temporal order, associate observations across viewpoints, or

Table 3: Results on RoboChrono. Performance across seven evaluation dimensions grouped into recognition, alignment, and grounding. Group scores are averages over their corresponding dimensions, and Choice Avg. is the mean accuracy over the six multiple-choice tasks.

Model	Size	Recognition					Alignment			Grounding	Overall
		Cur. Act.	Next Act.	Next +Goal	Frame Match	Group	View Match	Frame Order	Group	Action Time	
<i>chance floor</i>	–	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.130	0.250
GPT-6-Astra	–	0.894	0.764	0.836	0.983	0.873	0.675	0.683	0.679	0.721	0.805
RynnBrain1.1-122B-A10B	122B	0.752	0.511	0.588	0.954	0.701	0.536	0.329	0.442	0.512	0.612
Doubao-Seed-2.0-Lite	–	0.754	0.538	0.631	0.921	0.717	0.451	0.359	0.409	0.577	0.624
RynnBrain1.1-9B	9B	0.730	0.550	0.599	0.849	0.687	0.444	0.328	0.392	0.405	0.598
Qwen3-VL-235B-A22B	–	0.697	0.471	0.561	0.797	0.637	0.392	0.304	0.352	0.445	0.551
Qwen3-VL-32B	32B	0.709	0.509	0.566	0.720	0.630	0.307	0.329	0.317	0.469	0.536
Qwen3-VL-30B-A3B	30B-A3B	0.684	0.426	0.495	0.853	0.621	0.313	0.290	0.302	0.418	0.525
RynnBrain1.1-2B	2B	0.618	0.390	0.464	0.848	0.587	0.413	0.264	0.346	0.151	0.514
Qwen3-VL-8B	8B	0.653	0.459	0.501	0.690	0.580	0.324	0.284	0.306	0.385	0.498
Qwen3-VL-8B-Thinking	8B	0.608	0.453	0.508	0.715	0.575	0.315	0.277	0.298	0.335	0.492
MiMo-VL-7B-RL	7B	0.570	0.428	0.464	0.778	0.565	0.283	0.294	0.288	0.103	0.482
RynnBrain-2B	2B	0.574	0.377	0.438	0.656	0.516	0.344	0.254	0.303	0.188	0.452
Cosmos-Reason2-2B	2B	0.578	0.435	0.458	0.465	0.486	0.304	0.271	0.289	0.383	0.427
Gemini-3.6-Flash	–	0.284	0.308	0.347	0.245	0.295	0.608	0.496	0.558	0.187	0.374
Qwen3-VL-2B	2B	0.509	0.323	0.335	0.396	0.394	0.289	0.270	0.281	0.223	0.360
SenseNova-SI-1.1-2B	2B	0.409	0.349	0.415	0.398	0.393	0.266	0.286	0.275	0.001	0.358
Cosmos3-Edge-2B	2B	0.503	0.401	0.421	0.237	0.390	0.222	0.204	0.214	0.253	0.339
InternVL3-2B	2B	0.433	0.340	0.423	0.260	0.364	0.258	0.262	0.260	0.023	0.332

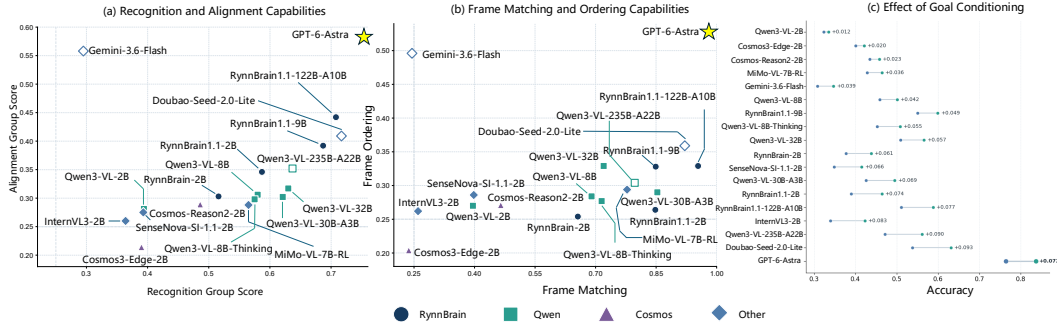


Figure 4: Diagnostic analysis of model capabilities on RoboChrono.

localize an action within the interaction timeline. Thus, even the strongest model retains substantial room for improvement on temporal progress understanding in real-world robot manipulation.

The remaining models exhibit much more uneven capability profiles. RynnBrain1.1-122B-A10B, for example, achieves 95.4% on Frame Matching but only 32.9% on Frame Ordering. This indicates that strong visual correspondence does not necessarily translate into understanding when events occur or in which order they unfold. Doubao-Seed-2.0-Lite shows a similar gap: it performs well on Current Action Recognition (75.4%) and Goal-Conditioned Next Action Prediction (63.1%), but reaches only 35.9% on Frame Ordering and 45.1% on View Matching. RynnBrain1.1-9B is comparatively strong on future prediction among the non-Astra models, obtaining 55.0% on Next Action Prediction, but its Frame Ordering accuracy remains only 32.8%. Together, these results suggest that predicting a plausible next action and recovering the temporal structure of an observed manipulation remain distinct challenges. Gemini-3.6-Flash exhibits a notably different profile. Its recognition performance is weak, with only 24.5% on Frame Matching and a Recognition Group score of 29.5%, yet it achieves the strongest non-Astra results on View Matching (60.8%) and Frame Ordering (49.6%). This contrast is visible in Figure 4(a,b) and shows that the benchmark separates recognition-oriented capability from alignment and temporal order reasoning rather than measuring a single monolithic notion of visual understanding, which aggregate accuracy alone would largely conceal.

Table 4: **Task-matched comparison across robot collection platforms.** We restrict GIM and Tianji gripper recordings to the 12 manipulation tasks available on both platforms and report question-weighted accuracy to isolate platform-specific effects.

Evaluation dimension	GIM	Tianji	Δ	Tianji > GIM (models)
Current Action	54.1	63.7	-9.6	17 / 18
Next Action	39.9	45.0	-5.1	14 / 18
Goal-conditioned Next Action	43.9	48.1	-4.2	14 / 18
Frame Matching	57.2	66.7	-9.6	17 / 18
View Matching	32.5	35.5	-3.0	15 / 18
Frame Ordering	27.1	30.0	-2.9	17 / 18
Choice Avg.	43.1	48.8	-5.8	18 / 18

At the lower end of the table, InternVL3-2B, Cosmos3-Edge-2B, SenseNova-SI-1.1-2B, and Qwen3-VL-2B obtain Choice Avg. scores between 33.2% and 36.0%. Their weaknesses are broad rather than task-specific. For example, InternVL3-2B and Cosmos3-Edge-2B remain close to the chance floor on several matching and ordering dimensions, while SenseNova-SI-1.1-2B achieves an Action Time score of only 0.001. These models therefore struggle not only with future prediction but also with basic temporal grounding and cross-observation reasoning.

Figure 4(c) further shows that explicit task goals consistently improve next-action prediction. Astra improves from 76.4% to 83.6%, while all other models also gain from goal conditioning. This indicates that knowing the intended task helps constrain plausible future actions. Nevertheless, the persistent gap between Next Action and the stronger recognition tasks shows that goal information alone is insufficient: models must still connect the current visual state with the temporal progress of the manipulation. Overall, the benchmark reveals a clear hierarchy of difficulty. Models are generally strongest at recognizing or matching visible observations, weaker at predicting future actions, and substantially less reliable at temporal ordering, cross-view alignment, and precise action-time grounding. GPT-6-Astra substantially raises the performance ceiling and shows that strong performance across all capability groups is achievable, but its remaining errors on ordering, viewpoint alignment, prediction, and temporal localization also indicate that real-world manipulation progress understanding is not yet solved, leaving substantial headroom for future models.

4.3 EFFECT OF ROBOT COLLECTION PLATFORM

To separate platform effects from task composition, we compare GIM and Tianji only on the 12 manipulation tasks recorded on both platforms.

As shown in Table 4 and Figure 5(a), model performance is broadly consistent across the two robot collection platforms. Tianji obtains moderately higher scores on all six matched dimensions, with an overall Choice Avg. difference of 5.8 percentage points, but the magnitude of the shift remains relatively small compared with the overall variation across models and tasks. The largest differences occur in Current Action Recognition and Frame Matching (~ 9.6 points), while Frame Ordering differs by only 2.9 points. Importantly, the overall capability pattern remains similar across platforms: dimensions that are difficult on GIM remain difficult on Tianji, and the relative strengths and weaknesses of the evaluated models are largely preserved. We therefore view the observed gap as a moderate sensitivity to collection conditions rather than evidence that benchmark performance is dominated by a particular robot platform or its specific camera and gripper hardware.

4.4 INPUT ABLATION

We further examine which input cues support benchmark performance using five open-weight models on a balanced 312-stem subset. Each ablation is evaluated against the Full condition on matched questions. Figure 5(b) reveals markedly different input dependencies across tasks. Removing visual observations causes large drops in Current Action Recognition and Frame Matching (-22.1 and -30.2 points), whereas Next Action Prediction changes by less than one point. Similarly, replacing the observation history with only the current frame reduces Current Action Recognition by 22.5 points but has only a small effect on future action prediction. This suggests that

Table 5: Input ablation on RoboChrono

Input	Current Action	Next Action	Goal-cond. Next	Frame Match	Frame Order	View Match	Action Time
Full	62.6	40.0	49.3	46.0	29.4	34.9	30.4
Blind	40.5 (-22.1)	39.3 (-0.7)	41.1 (-8.2)	15.8 (-30.2)	27.5 (-1.9)	22.8 (-12.1)	7.6 (-22.8)
Text Only	—	—	—	27.6 (-18.4)	—	24.8 (-10.1)	—
Current Frame Only	40.0 (-22.5)	37.5 (-2.6)	46.2 (-3.0)	—	—	—	—
Shuffled Frames	53.6 (-8.9)	39.2 (-0.9)	47.6 (-1.7)	56.4 (+10.4)	—	—	12.1 (-18.3)
Identity Re-encode	62.1 (-0.4)	—	—	45.9 (+0.0)	—	—	—

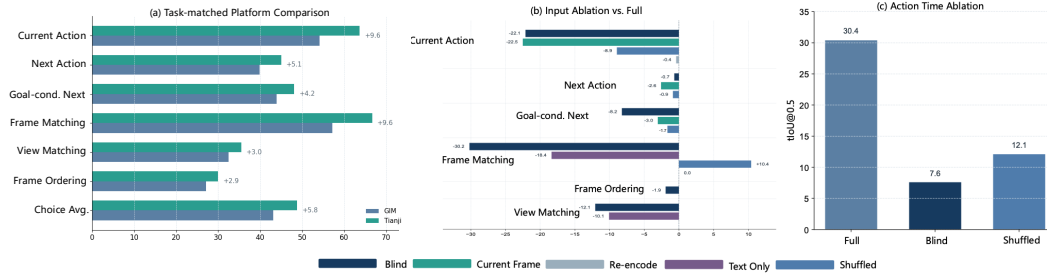


Figure 5: Controlled analysis of collection and input effects.

current action understanding relies strongly on visual history, while next-action prediction can often be supported by task and action priors even when visual evidence is entirely unavailable.

Temporal order produces a complementary effect. Shuffling frames reduces Current Action Recognition by 8.9 points and Action Time Localization from 30.4 to 12.1 (Figure 5(c)), showing that temporal structure is important for grounding task progress. By contrast, Frame Matching increases under shuffling, while the identity re-encoding control produces almost no change. The effect therefore reflects task specific use of temporal order rather than a generic video processing artifact.

Overall, the ablations expose a clear distinction between *recognizing visible evidence* and *understanding task progression*. Visual observations are essential for recognition and matching, temporal context is critical for current state and temporal grounding, while next-action prediction remains comparatively insensitive to the exact visual history, relying more on task priors.

5 CONCLUSION

We introduced RoboChrono, a real-robot benchmark for evaluating temporal task understanding from progressively revealed visual observations, covering 39 manipulation scenarios and seven complementary evaluation tasks. Our results reveal a persistent gap between recognizing visible content and understanding how a physical task unfolds over time. Even strong multimodal models exhibit substantial weaknesses in temporal ordering, future action prediction, viewpoint alignment, and temporal localization, while controlled ablations further reveal distinct dependencies on observation history, temporal order, and task priors. These findings highlight the importance of evaluating temporal capabilities individually rather than relying solely on aggregate performance. We hope RoboChrono provides a useful testbed for developing multimodal systems that can more reliably reason about task progress during ongoing real-world robot interaction.

AI USE STATEMENT

Generative AI tools were used only for language editing and writing assistance. They were not used for data generation, experimental execution, result analysis, or scientific decision making. All AI-assisted content was reviewed by the authors, who take full responsibility for the final manuscript.

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A APPENDIX

Hardware Platforms. Our benchmark is built upon two real-world robotic manipulation platforms, covering different robot embodiments and workspace configurations. As illustrated in Fig. 6, the first platform is a dual-arm robotic system for whole-workspace manipulation, while the second platform consists of two tabletop robotic manipulators. Both platforms are operated through a VR-based teleoperation interface, where human operators use a head-mounted display and handheld controllers to perform manipulation tasks. These complementary hardware setups provide diverse real-world robot trajectories for benchmark construction and evaluation.



Figure 6: **Hardware platforms used in our benchmark.** We employ two real-world robotic manipulation setups with different embodiments and workspace configurations. Both platforms support VR-based human teleoperation for collecting manipulation trajectories.

Conditional Action Time Evaluation. The Action Time task requires models to answer with temporal intervals in seconds. Although the corrected prompt states the total duration of the queried video, some systems still return intervals as fractions of the video duration. This behavior is a unit error rather than a temporal localization error. We therefore report a conditional Action Time score that evaluates only the answers that are given in seconds.

Let S denote the set of localization queries for which the model returns a parseable interval in seconds. The conditional score is computed as

$$\frac{1}{|S|} \sum_{i \in S} \mathbf{1} [\text{tIoU}(\hat{I}_i, I_i^*) \geq 0.5].$$

The reference interval and scoring threshold are unchanged from the main evaluation. The only difference is that normalized interval answers are excluded from the denominator.

Qualitative Action Time Examples. Figure 7 presents six qualitative examples of Action Time localization from the GIM and Tianji platforms, comparing the predictions of five models with the ground-truth intervals. The examples illustrate both accurate temporal alignment and different failure cases, including overly broad intervals, predictions that do not overlap the reference interval, and timestamps beyond the video duration.

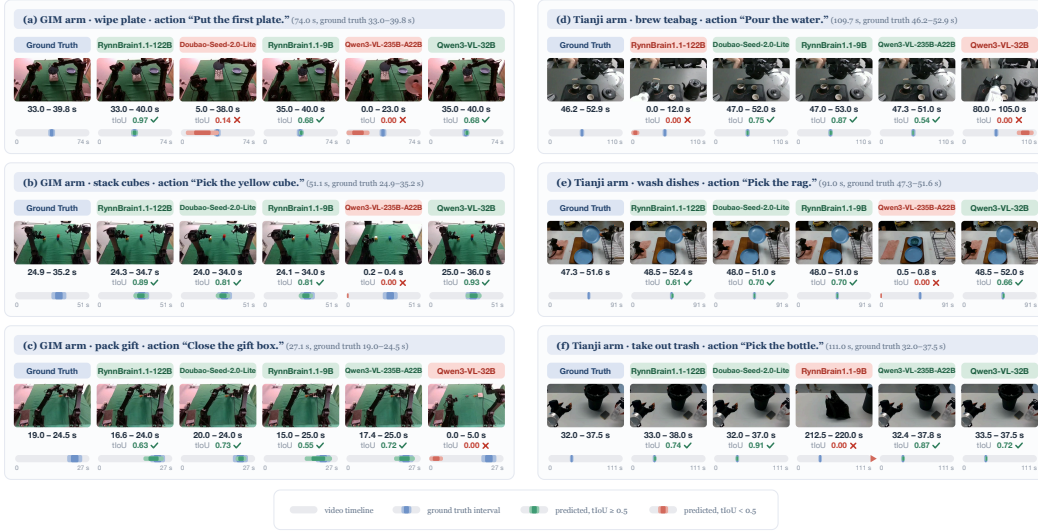


Figure 7: Qualitative examples of Action Time localization.

RoboChrono Evaluation Pipeline. Algorithm 1 summarizes the complete RoboChrono evaluation pipeline across different models, scenarios, and evaluation dimensions. For each question, the pipeline prepares the visual input, performs model inference, parses either a multiple-choice answer or a temporal interval, and computes the corresponding accuracy or tIoU@0.5 score. Finally, all question-level results are aggregated into the evaluation report.

Algorithm 1: RoboChrono Evaluation

Input: Models $\mathcal{M}$, scenarios $\mathcal{S}$, dimensions $\mathcal{D}$

Output: Evaluation report $\mathcal{R}$

```

1 Validate dataset and model configurations;
2 Initialize result set  $\mathcal{R} \leftarrow \emptyset$ ;
3 foreach model  $m \in \mathcal{M}$  do
4   Initialize model adapter  $A_m$ ;
5   foreach scenario  $s \in \mathcal{S}$  do
6     foreach dimension  $d \in \mathcal{D}$  do
7       Load questions  $Q_{s,d}$ ;
8       foreach question  $q \in Q_{s,d}$  do
9         if  $q$  has already been completed then
10           continue;
11          $x \leftarrow \text{PrepareInput}(q)$ ;
12          $y \leftarrow A_m.\text{Generate}(x)$ ;
13         if  $d = \text{action.time}$  then
14            $\hat{t} \leftarrow \text{ParseInterval}(y)$ ;
15            $z \leftarrow \mathbb{I}(\text{tIoU}(\hat{t}, t_q) \geq 0.5)$ ;
16         else
17            $\hat{c} \leftarrow \text{ParseChoice}(y)$ ;
18            $z \leftarrow \mathbb{I}(\hat{c} = c_q)$ ;
19         Save  $(m, s, d, q, y, z)$ ;
20        $\text{Score}_{m,s,d} \leftarrow \frac{1}{|Q_{s,d}|} \sum_{q \in Q_{s,d}} z_q$ ;
21 Aggregate all scores into  $\mathcal{R}$ ;
22 return  $\mathcal{R}$ ;

```

B COMPLETE SCENARIO-LEVEL EVALUATION DETAILS

The following tables present the scenario-level data for each model across 39 scenarios and seven tasks.

Table 6: Complete scenario-level evaluation details for RynnBrain1.1-122B-A10B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.967	0.561	0.780	0.943	0.659	0.366	0.480
box_pen_hand	0.875	0.292	0.300	1.000	–	0.388	0.883
box_pen_tianjihand	0.513	0.300	0.253	0.993	0.520	0.430	0.700
box_shoe_tianji	0.933	1.000	1.000	0.940	0.672	0.462	0.387
box_shoes_gim	0.952	1.000	1.000	0.973	0.646	0.388	0.585
brew_teabag_gim	0.712	0.364	0.439	0.902	0.538	0.333	0.189
brew_teabag_tianji	0.647	0.506	0.449	1.000	0.474	0.340	0.692
cap_pen_gim	0.639	0.328	0.639	0.989	0.328	0.350	0.728
make_tea_gim	0.641	0.282	0.096	0.968	0.436	0.237	0.538
make_tea_tianji	1.000	0.376	0.615	0.949	0.564	0.291	0.940
move_flower_hand	0.913	–	–	–	–	–	0.938
move_flower_tianjihand	0.700	0.767	0.900	0.967	0.678	0.500	0.367
move_gift_tianjihand	0.567	0.983	0.967	0.967	0.533	0.333	0.189
pack_aidkit_gim	0.800	0.593	0.607	0.913	0.693	0.340	0.287
pack_aidkit_tianji	0.750	0.742	0.783	0.933	0.367	0.392	0.117
pack_airpods_gim	0.667	0.370	0.333	0.926	0.500	0.278	0.574
pack_airpods_tianji	0.708	0.333	0.492	0.992	0.275	0.258	0.100
pack_express_tianji	0.667	0.293	0.280	0.980	0.327	0.250	0.520
pack_gift_gim	0.527	0.660	0.640	0.893	0.687	0.300	0.293
pack_gift_hand	0.692	0.342	0.400	0.967	–	0.375	0.350
pack_gift_tianji	0.561	0.122	0.244	0.911	0.634	0.415	0.236
pack_sunglasses_hand	0.748	0.724	0.919	0.927	–	0.407	0.602
slip_tshirt_gim	0.681	0.521	0.563	0.847	0.299	0.448	0.285
sort_cubes_gim	0.802	0.921	0.966	0.915	0.695	0.288	0.486
sort_cubes_tianji	0.821	0.764	0.878	1.000	0.691	0.317	0.439
stack_cubes_gim	0.700	0.680	0.767	0.980	0.153	0.353	0.587
stack_cubes_hand	0.977	0.581	0.682	0.984	–	0.407	0.667
stack_cubes_tianji	1.000	0.727	0.807	1.000	0.827	0.300	0.780
stack_cubes_tianjihand	1.000	0.600	0.713	0.967	0.747	0.400	0.900
stow_sunglasses_tianjihand	0.613	0.510	0.500	0.987	0.573	0.360	0.720
takeout_trash_tianji	0.957	0.812	0.795	0.983	0.248	0.188	0.855
tidy_stationery_gim	0.722	0.301	0.611	0.954	0.543	0.259	0.605
tidy_stationery_tianji	0.587	0.476	0.561	1.000	0.561	0.429	0.598
wash_dishes_gim	0.884	0.415	0.503	0.925	0.755	0.299	0.265
wash_dishes_tianji	0.892	0.533	0.508	0.842	0.650	0.200	0.383
wipe_plate_gim	0.673	0.307	0.227	0.880	0.360	0.293	0.333
wipe_plate_tianji	0.575	0.250	0.408	1.000	0.642	0.250	0.050
zip_pouch_gim	0.806	0.396	0.615	0.986	0.396	0.333	0.500
zip_pouch_tianji	0.654	0.353	0.686	0.948	0.575	0.333	0.503
Pooled (valid)	0.752	0.511	0.588	0.954	0.536	0.329	0.512
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 7: Complete scenario-level evaluation details for RynnBrain1.1-9B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.829	0.366	0.659	0.780	0.455	0.146	0.593
box_pen_hand	0.783	0.458	0.600	0.958	–	0.300	0.808
box_pen_tianjihand	0.453	0.327	0.493	0.980	0.413	0.420	0.427
box_shoe_tianji	0.992	1.000	1.000	0.718	0.546	0.308	0.613
box_shoes_gim	0.980	1.000	1.000	0.701	0.449	0.347	0.735
brew_teabag_gim	0.636	0.364	0.500	0.886	0.417	0.379	0.068
brew_teabag_tianji	0.667	0.442	0.359	1.000	0.385	0.346	0.353
cap_pen_gim	0.550	0.328	0.439	0.967	0.311	0.325	0.233
make_tea_gim	0.513	0.250	0.032	0.974	0.359	0.224	0.244
make_tea_tianji	0.974	0.615	0.479	0.795	0.453	0.316	0.812
move_flower_hand	1.000	–	–	–	–	–	0.900
move_flower_tianjihand	0.767	0.633	0.383	0.678	0.544	0.433	0.344
move_gift_tianjihand	0.489	0.650	0.900	0.667	0.500	0.400	0.211
pack_aidkit_gim	0.820	0.487	0.547	0.767	0.627	0.387	0.260
pack_aidkit_tianji	0.650	0.742	0.758	0.883	0.367	0.425	0.100
pack_airpods_gim	0.611	0.222	0.315	0.852	0.315	0.352	0.167
pack_airpods_tianji	0.683	0.375	0.283	0.992	0.317	0.275	0.050
pack_express_tianji	0.593	0.287	0.380	0.727	0.420	0.310	0.533
pack_gift_gim	0.540	0.733	0.640	0.800	0.540	0.287	0.140
pack_gift_hand	0.567	0.342	0.458	0.967	–	0.417	0.267
pack_gift_tianji	0.756	0.309	0.154	0.886	0.528	0.447	0.528
pack_sunglasses_hand	0.780	0.854	0.984	0.683	–	0.390	0.528
slip_tshirt_gim	0.639	0.521	0.938	0.736	0.264	0.323	0.424
sort_cubes_gim	0.859	0.881	0.904	0.751	0.531	0.322	0.305
sort_cubes_tianji	0.935	0.935	0.951	0.976	0.496	0.301	0.309
stack_cubes_gim	0.673	0.667	0.660	0.947	0.280	0.400	0.507
stack_cubes_hand	0.930	0.860	0.922	0.984	–	0.430	0.659
stack_cubes_tianji	0.987	0.847	0.893	0.820	0.507	0.270	0.593
stack_cubes_tianjihand	0.987	0.827	0.907	0.960	0.647	0.260	0.700
stow_sunglasses_tianjihand	0.473	0.760	0.670	0.980	0.547	0.320	0.533
takeout_trash_tianji	0.949	0.821	0.812	0.872	0.436	0.171	0.291
tidy_stationery_gim	0.667	0.593	0.778	0.759	0.380	0.269	0.432
tidy_stationery_tianji	0.481	0.487	0.619	0.989	0.381	0.508	–
wash_dishes_gim	0.878	0.401	0.429	0.762	0.544	0.279	0.211
wash_dishes_tianji	0.858	0.342	0.392	0.742	0.575	0.333	0.242
wipe_plate_gim	0.693	0.273	0.200	0.647	0.373	0.247	0.473
wipe_plate_tianji	0.675	0.200	0.275	0.925	0.458	0.233	0.083
zip_pouch_gim	0.701	0.406	0.646	0.799	0.361	0.354	0.486
zip_pouch_tianji	0.797	0.471	0.529	0.908	0.471	0.196	0.216
Pooled (valid)	0.730	0.550	0.599	0.849	0.444	0.328	0.405
Valid record count	5407	4880	4880	5325	4835	3979	5218

Table 8: Complete scenario-level evaluation details for RynnBrain1.1-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.634	0.146	0.500	0.756	0.317	0.244	0.220
box_pen_hand	0.775	0.442	0.558	0.967	–	0.263	0.375
box_pen_tianjihand	0.433	0.333	0.540	0.953	0.393	0.330	0.080
box_shoe_tianji	0.992	0.861	0.848	0.812	0.529	0.308	0.303
box_shoes_gim	0.946	0.643	0.684	0.769	0.435	0.245	0.388
brew_teabag_gim	0.356	0.402	0.341	0.833	0.477	0.265	0.000
brew_teabag_tianji	0.622	0.385	0.410	0.955	0.410	0.308	0.199
cap_pen_gim	0.417	0.306	0.439	0.894	0.339	0.325	0.050
make_tea_gim	0.282	0.135	0.231	0.891	0.276	0.237	0.000
make_tea_tianji	0.897	0.479	0.530	0.812	0.402	0.325	0.205
move_flower_hand	0.825	–	–	–	–	–	0.400
move_flower_tianjihand	0.567	0.717	0.767	0.833	0.389	0.400	0.167
move_gift_tianjihand	0.300	0.350	0.633	0.767	0.400	0.300	0.000
pack_aidkit_gim	0.667	0.453	0.480	0.700	0.553	0.273	0.013
pack_aidkit_tianji	0.617	0.558	0.600	0.658	0.350	0.383	0.058
pack_airpods_gim	0.315	0.222	0.278	0.944	0.519	0.315	0.074
pack_airpods_tianji	0.475	0.300	0.342	0.958	0.283	0.225	0.008
pack_express_tianji	0.347	0.193	0.487	0.873	0.247	0.230	0.307
pack_gift_gim	0.473	0.353	0.360	0.740	0.400	0.253	0.027
pack_gift_hand	0.442	0.233	0.317	0.900	–	0.342	0.042
pack_gift_tianji	0.634	0.252	0.179	0.805	0.650	0.358	0.049
pack_sunglasses_hand	0.870	0.447	0.480	0.846	–	0.301	0.073
slip_tshirt_gim	0.569	0.382	0.632	0.847	0.278	0.219	0.264
sort_cubes_gim	0.740	0.627	0.599	0.780	0.486	0.220	0.073
sort_cubes_tianji	0.821	0.626	0.724	0.967	0.512	0.244	0.333
stack_cubes_gim	0.620	0.473	0.473	0.827	0.273	0.240	0.073
stack_cubes_hand	0.791	0.364	0.558	0.969	–	0.384	0.419
stack_cubes_tianji	0.973	0.560	0.573	0.907	0.467	0.220	0.400
stack_cubes_tianjihand	0.873	0.300	0.467	0.953	0.600	0.220	0.333
stow_sunglasses_tianjihand	0.507	0.530	0.560	0.907	0.440	0.300	0.333
takeout_trash_tianji	0.880	0.564	0.530	0.915	0.368	0.162	0.145
tidy_stationery_gim	0.466	0.241	0.514	0.843	0.383	0.204	0.056
tidy_stationery_tianji	0.370	0.407	0.302	0.979	0.386	0.317	0.196
wash_dishes_gim	0.735	0.252	0.272	0.782	0.578	0.238	0.020
wash_dishes_tianji	0.750	0.417	0.433	0.775	0.458	0.142	0.192
wipe_plate_gim	0.553	0.347	0.273	0.640	0.353	0.173	0.013
wipe_plate_tianji	0.625	0.258	0.225	0.667	0.517	0.217	0.025
zip_pouch_gim	0.597	0.260	0.448	0.813	0.361	0.333	0.083
zip_pouch_tianji	0.667	0.353	0.490	0.935	0.379	0.235	0.072
Pooled (valid)	0.618	0.390	0.464	0.848	0.413	0.264	0.151
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 9: Complete scenario-level evaluation details for RynnBrain-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.447	0.024	0.476	0.528	0.301	0.195	0.073
box_pen_hand	0.842	0.217	0.158	0.983	–	0.238	0.308
box_pen_tianjihand	0.560	0.120	0.093	0.933	0.347	0.240	0.133
box_shoe_tianji	0.992	0.987	0.975	0.658	0.378	0.308	0.311
box_shoes_gim	0.973	0.878	0.959	0.605	0.401	0.306	0.306
brew_teabag_gim	0.447	0.545	0.409	0.614	0.303	0.250	0.015
brew_teabag_tianji	0.481	0.397	0.276	0.929	0.417	0.250	0.250
cap_pen_gim	0.422	0.233	0.278	0.578	0.294	0.242	0.128
make_tea_gim	0.353	0.160	0.199	0.692	0.321	0.250	0.045
make_tea_tianji	0.880	0.299	0.299	0.692	0.385	0.333	0.179
move_flower_hand	0.825	–	–	–	–	–	0.400
move_flower_tianjihand	0.500	0.450	0.483	0.456	0.367	0.333	0.278
move_gift_tianjihand	0.278	0.683	0.900	0.589	0.344	0.333	0.122
pack_aidkit_gim	0.460	0.273	0.327	0.567	0.387	0.207	0.013
pack_aidkit_tianji	0.592	0.483	0.483	0.683	0.375	0.367	0.033
pack_airpods_gim	0.389	0.259	0.333	0.537	0.315	0.278	0.148
pack_airpods_tianji	0.517	0.225	0.292	0.725	0.250	0.233	0.017
pack_express_tianji	0.373	0.100	0.340	0.833	0.193	0.300	0.247
pack_gift_gim	0.287	0.307	0.307	0.400	0.273	0.307	0.027
pack_gift_hand	0.308	0.250	0.333	0.858	–	0.333	0.200
pack_gift_tianji	0.626	0.154	0.138	0.724	0.317	0.366	0.106
pack_sunglasses_hand	0.764	0.553	0.382	0.724	–	0.293	0.341
slip_tshirt_gim	0.528	0.507	0.583	0.424	0.299	0.219	0.306
sort_cubes_gim	0.746	0.554	0.723	0.322	0.322	0.186	0.158
sort_cubes_tianji	0.707	0.821	0.748	0.772	0.455	0.293	0.203
stack_cubes_gim	0.500	0.413	0.467	0.607	0.340	0.213	0.120
stack_cubes_hand	0.783	0.558	0.721	0.814	–	0.221	0.395
stack_cubes_tianji	0.987	0.647	0.667	0.760	0.347	0.280	0.613
stack_cubes_tianjihand	0.853	0.327	0.640	0.960	0.447	0.220	0.340
stow_sunglasses_tianjihand	0.287	0.680	0.670	0.807	0.433	0.300	0.187
takeout_trash_tianji	0.906	0.744	0.675	0.915	0.402	0.145	0.128
tidy_stationery_gim	0.435	0.218	0.486	0.417	0.346	0.194	0.120
tidy_stationery_tianji	0.302	0.508	0.672	0.804	0.429	0.389	0.307
wash_dishes_gim	0.646	0.333	0.293	0.388	0.347	0.177	0.136
wash_dishes_tianji	0.642	0.442	0.483	0.642	0.350	0.175	0.067
wipe_plate_gim	0.507	0.160	0.087	0.407	0.280	0.173	0.020
wipe_plate_tianji	0.608	0.100	0.175	0.508	0.375	0.225	0.050
zip_pouch_gim	0.521	0.135	0.396	0.583	0.257	0.250	0.146
zip_pouch_tianji	0.510	0.049	0.235	0.778	0.314	0.255	0.412
Pooled (valid)	0.574	0.377	0.438	0.656	0.344	0.254	0.188
Valid record count	5407	4880	4880	5325	4835	3979	5407

Coverage: Full question set. Choice Avg (pooled, valid): 0.452 (29306 questions). --: no evaluated questions. The task and Choice Avg denominators exclude source-fault cells.

Table 10: Complete scenario-level evaluation details for Qwen3-VL-32B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.886	0.354	0.549	0.854	0.244	0.317	0.512
box_pen_hand	0.833	0.550	0.775	0.842	–	0.225	0.658
box_pen_tianjihand	0.420	0.347	0.407	0.973	0.387	0.290	0.567
box_shoe_tianji	1.000	1.000	1.000	0.667	0.303	0.436	0.403
box_shoes_gim	0.993	0.990	0.949	0.694	0.347	0.388	0.476
brew_teabag_gim	0.598	0.598	0.523	0.803	0.174	0.288	0.326
brew_teabag_tianji	0.686	0.494	0.474	0.987	0.276	0.378	0.763
cap_pen_gim	0.656	0.228	0.456	0.911	0.194	0.417	0.606
make_tea_gim	0.404	0.224	0.186	0.917	0.179	0.256	0.340
make_tea_tianji	1.000	0.368	0.487	0.915	0.376	0.291	0.966
move_flower_hand	0.925	–	–	–	–	–	0.675
move_flower_tianjihand	0.756	0.633	0.683	0.367	0.344	0.600	0.322
move_gift_tianjihand	0.322	0.883	0.933	0.622	0.311	0.433	0.167
pack_aidkit_gim	0.807	0.540	0.580	0.647	0.407	0.387	0.347
pack_aidkit_tianji	0.783	0.783	0.800	0.700	0.350	0.342	0.183
pack_airpods_gim	0.667	0.315	0.407	0.741	0.370	0.426	0.407
pack_airpods_tianji	0.558	0.317	0.317	0.875	0.133	0.250	0.217
pack_express_tianji	0.607	0.307	0.313	0.920	0.173	0.260	0.593
pack_gift_gim	0.353	0.547	0.433	0.513	0.233	0.320	0.120
pack_gift_hand	0.350	0.308	0.300	0.667	–	0.325	0.258
pack_gift_tianji	0.683	0.211	0.260	0.650	0.341	0.407	0.439
pack_sunglasses_hand	0.748	0.626	0.821	0.642	–	0.374	0.236
slip_tshirt_gim	0.840	0.472	0.743	0.688	0.333	0.406	0.389
sort_cubes_gim	0.768	0.785	0.825	0.486	0.441	0.339	0.469
sort_cubes_tianji	0.902	0.894	0.943	0.813	0.439	0.317	0.350
stack_cubes_gim	0.833	0.533	0.627	0.513	0.313	0.327	0.800
stack_cubes_hand	0.984	0.721	0.806	0.550	–	0.430	0.698
stack_cubes_tianji	1.000	0.780	0.920	0.487	0.380	0.350	0.973
stack_cubes_tianjihand	1.000	0.800	0.933	0.740	0.480	0.370	0.753
stow_sunglasses_tianjihand	0.587	0.570	0.500	0.700	0.313	0.320	0.600
takeout_trash_tianji	0.949	0.821	0.872	0.735	0.350	0.162	0.333
tidy_stationery_gim	0.660	0.296	0.366	0.688	0.269	0.250	0.485
tidy_stationery_tianji	0.444	0.556	0.587	0.762	0.243	0.413	0.376
wash_dishes_gim	0.728	0.252	0.327	0.653	0.374	0.259	0.381
wash_dishes_tianji	0.758	0.325	0.417	0.600	0.450	0.325	0.408
wipe_plate_gim	0.713	0.387	0.313	0.573	0.267	0.327	0.400
wipe_plate_tianji	0.642	0.258	0.283	0.667	0.375	0.225	0.192
zip_pouch_gim	0.486	0.385	0.438	0.833	0.236	0.458	0.458
zip_pouch_tianji	0.614	0.451	0.510	0.856	0.190	0.294	0.346
Pooled (valid)	0.709	0.509	0.566	0.720	0.307	0.329	0.469
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 11: Complete scenario-level evaluation details for Qwen3-VL-30B-A3B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.797	0.049	0.366	0.821	0.285	0.390	0.528
box_pen_hand	0.900	0.458	0.417	0.983	–	0.325	0.858
box_pen_tianjihand	0.513	0.273	0.300	0.980	0.393	0.290	0.527
box_shoe_tianji	1.000	0.987	0.987	0.658	0.471	0.333	0.429
box_shoes_gim	0.980	0.878	0.776	0.782	0.361	0.286	0.551
brew_teabag_gim	0.682	0.152	0.114	0.742	0.212	0.280	0.235
brew_teabag_tianji	0.622	0.417	0.455	0.987	0.237	0.346	0.641
cap_pen_gim	0.400	0.311	0.517	0.972	0.189	0.317	0.350
make_tea_gim	0.378	0.154	0.160	0.936	0.179	0.295	0.199
make_tea_tianji	1.000	0.214	0.179	0.923	0.385	0.282	0.974
move_flower_hand	0.988	–	–	–	–	–	0.850
move_flower_tianjihand	0.700	0.817	0.983	0.411	0.289	0.400	0.267
move_gift_tianjihand	0.200	0.817	0.783	0.700	0.400	0.433	0.133
pack_aidkit_gim	0.687	0.353	0.260	0.727	0.400	0.307	0.220
pack_aidkit_tianji	0.675	0.433	0.433	0.700	0.425	0.308	0.217
pack_airpods_gim	0.648	0.167	0.315	0.907	0.222	0.315	0.259
pack_airpods_tianji	0.608	0.183	0.250	0.983	0.250	0.275	0.050
pack_express_tianji	0.647	0.127	0.293	0.927	0.193	0.300	0.173
pack_gift_gim	0.340	0.347	0.393	0.773	0.247	0.313	0.060
pack_gift_hand	0.467	0.308	0.333	0.908	–	0.342	0.275
pack_gift_tianji	0.675	0.203	0.317	0.870	0.333	0.390	0.309
pack_sunglasses_hand	0.854	0.691	0.829	0.740	–	0.325	0.650
slip_tshirt_gim	0.701	0.444	0.556	0.819	0.306	0.167	0.375
sort_cubes_gim	0.627	0.763	0.802	0.729	0.316	0.277	0.243
sort_cubes_tianji	0.927	0.878	0.870	0.984	0.398	0.268	0.333
stack_cubes_gim	0.700	0.620	0.607	0.907	0.300	0.293	0.533
stack_cubes_hand	0.984	0.643	0.721	0.930	–	0.360	0.698
stack_cubes_tianji	1.000	0.713	0.733	0.893	0.327	0.220	0.807
stack_cubes_tianjihand	1.000	0.667	0.767	1.000	0.507	0.190	0.667
stow_sunglasses_tianjihand	0.540	0.590	0.820	0.980	0.307	0.300	0.593
takeout_trash_tianji	0.906	0.829	0.744	0.923	0.402	0.162	0.342
tidy_stationery_gim	0.546	0.190	0.338	0.833	0.262	0.287	0.290
tidy_stationery_tianji	0.519	0.566	0.725	1.000	0.265	0.333	0.376
wash_dishes_gim	0.816	0.238	0.320	0.803	0.333	0.279	0.354
wash_dishes_tianji	0.825	0.475	0.483	0.833	0.333	0.267	0.425
wipe_plate_gim	0.587	0.300	0.280	0.653	0.280	0.233	0.407
wipe_plate_tianji	0.700	0.158	0.192	0.667	0.375	0.192	0.108
zip_pouch_gim	0.396	0.083	0.479	0.813	0.257	0.500	0.472
zip_pouch_tianji	0.673	0.127	0.510	0.967	0.353	0.157	0.686
Pooled (valid)	0.684	0.426	0.495	0.853	0.313	0.290	0.418
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 12: Complete scenario-level evaluation details for Qwen3-VL-8B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.805	0.488	0.500	0.732	0.301	0.195	0.650
box_pen_hand	0.750	0.175	0.150	0.883	–	0.313	0.550
box_pen_tianjihand	0.553	0.247	0.213	0.960	0.440	0.260	0.560
box_shoe_tianji	0.992	1.000	1.000	0.496	0.412	0.410	0.361
box_shoes_gim	0.980	0.990	1.000	0.531	0.333	0.286	0.578
brew_teabag_gim	0.439	0.341	0.356	0.621	0.273	0.295	0.227
brew_teabag_tianji	0.577	0.506	0.474	0.936	0.308	0.333	0.545
cap_pen_gim	0.572	0.322	0.406	0.872	0.200	0.342	0.356
make_tea_gim	0.385	0.173	0.154	0.788	0.269	0.282	0.167
make_tea_tianji	0.983	0.299	0.162	0.778	0.419	0.291	0.872
move_flower_hand	0.925	–	–	–	–	–	0.450
move_flower_tianjihand	0.756	0.700	0.750	0.133	0.378	0.367	0.267
move_gift_tianjihand	0.300	0.683	0.767	0.678	0.322	0.267	0.167
pack_aidkit_gim	0.793	0.507	0.507	0.633	0.440	0.293	0.100
pack_aidkit_tianji	0.700	0.742	0.750	0.658	0.358	0.342	0.250
pack_airpods_gim	0.481	0.370	0.296	0.778	0.278	0.333	0.315
pack_airpods_tianji	0.525	0.233	0.275	0.925	0.250	0.267	0.025
pack_express_tianji	0.573	0.207	0.260	0.713	0.260	0.280	0.373
pack_gift_gim	0.233	0.533	0.407	0.673	0.280	0.280	0.027
pack_gift_hand	0.333	0.117	0.325	0.767	–	0.325	0.183
pack_gift_tianji	0.415	0.382	0.390	0.724	0.350	0.382	0.358
pack_sunglasses_hand	0.707	0.593	0.593	0.577	–	0.350	0.423
slip_tshirt_gim	0.479	0.306	0.576	0.688	0.319	0.250	0.333
sort_cubes_gim	0.723	0.763	0.797	0.492	0.356	0.266	0.316
sort_cubes_tianji	0.911	0.919	0.911	0.756	0.407	0.285	0.122
stack_cubes_gim	0.767	0.560	0.527	0.687	0.320	0.253	0.587
stack_cubes_hand	1.000	0.620	0.667	0.806	–	0.326	0.589
stack_cubes_tianji	1.000	0.660	0.667	0.847	0.353	0.220	0.953
stack_cubes_tianjihand	0.973	0.467	0.773	0.920	0.493	0.240	0.667
stow_sunglasses_tianjihand	0.380	0.430	0.630	0.787	0.347	0.400	0.507
takeout_trash_tianji	0.932	0.769	0.838	0.684	0.325	0.188	0.214
tidy_stationery_gim	0.667	0.477	0.389	0.605	0.256	0.213	0.444
tidy_stationery_tianji	0.466	0.503	0.677	0.979	0.333	0.349	0.243
wash_dishes_gim	0.714	0.218	0.286	0.415	0.306	0.252	0.374
wash_dishes_tianji	0.725	0.425	0.475	0.517	0.283	0.208	0.367
wipe_plate_gim	0.540	0.267	0.193	0.413	0.267	0.220	0.207
wipe_plate_tianji	0.667	0.192	0.258	0.333	0.317	0.233	0.275
zip_pouch_gim	0.438	0.323	0.469	0.590	0.292	0.396	0.361
zip_pouch_tianji	0.497	0.461	0.784	0.634	0.281	0.196	0.451
Pooled (valid)	0.653	0.459	0.501	0.690	0.324	0.284	0.385
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 13: Complete scenario-level evaluation details for Qwen3-VL-8B-Thinking.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.846	0.524	0.537	0.683	0.358	0.390	0.309
box_pen_hand	0.700	0.467	0.408	0.933	–	0.300	0.642
box_pen_tianjihand	0.480	0.327	0.307	0.913	0.347	0.280	0.467
box_shoe_tianji	0.941	0.823	0.835	0.718	0.370	0.564	0.420
box_shoes_gim	0.966	0.857	0.959	0.633	0.388	0.347	0.483
brew_teabag_gim	0.348	0.379	0.470	0.720	0.258	0.265	0.098
brew_teabag_tianji	0.679	0.288	0.359	0.929	0.263	0.308	0.333
cap_pen_gim	0.428	0.394	0.428	0.833	0.194	0.275	0.317
make_tea_gim	0.256	0.186	0.288	0.827	0.269	0.276	0.103
make_tea_tianji	0.872	0.402	0.222	0.795	0.393	0.291	0.786
move_flower_hand	0.838	–	–	–	–	–	0.675
move_flower_tianjihand	0.689	0.650	0.450	0.311	0.367	0.300	0.300
move_gift_tianjihand	0.400	0.600	0.683	0.622	0.344	0.267	0.178
pack_aidkit_gim	0.627	0.400	0.427	0.633	0.407	0.267	0.080
pack_aidkit_tianji	0.617	0.500	0.575	0.492	0.325	0.325	0.150
pack_airpods_gim	0.519	0.352	0.333	0.833	0.370	0.315	0.296
pack_airpods_tianji	0.575	0.175	0.208	0.967	0.175	0.233	0.050
pack_express_tianji	0.500	0.380	0.413	0.773	0.233	0.320	0.520
pack_gift_gim	0.280	0.453	0.400	0.620	0.253	0.233	0.047
pack_gift_hand	0.383	0.250	0.358	0.750	–	0.308	0.250
pack_gift_tianji	0.415	0.407	0.350	0.593	0.268	0.317	0.220
pack_sunglasses_hand	0.764	0.553	0.455	0.569	–	0.301	0.423
slip_tshirt_gim	0.556	0.403	0.757	0.632	0.257	0.292	0.535
sort_cubes_gim	0.655	0.746	0.695	0.605	0.390	0.226	0.237
sort_cubes_tianji	0.683	0.878	0.886	0.846	0.382	0.260	0.146
stack_cubes_gim	0.667	0.567	0.513	0.700	0.193	0.220	0.480
stack_cubes_hand	0.822	0.574	0.760	0.837	–	0.395	0.566
stack_cubes_tianji	0.987	0.600	0.693	0.647	0.473	0.290	0.740
stack_cubes_tianjihand	0.980	0.560	0.687	0.827	0.507	0.270	0.400
stow_sunglasses_tianjihand	0.447	0.680	0.880	0.833	0.433	0.320	0.420
takeout_trash_tianji	0.744	0.598	0.632	0.812	0.265	0.197	0.350
tidy_stationery_gim	0.571	0.375	0.606	0.590	0.228	0.241	0.262
tidy_stationery_tianji	0.444	0.450	0.450	0.963	0.217	0.333	0.265
wash_dishes_gim	0.646	0.279	0.333	0.537	0.429	0.238	0.340
wash_dishes_tianji	0.550	0.350	0.367	0.708	0.400	0.200	0.383
wipe_plate_gim	0.507	0.187	0.153	0.547	0.233	0.207	0.220
wipe_plate_tianji	0.617	0.233	0.233	0.433	0.358	0.258	0.267
zip_pouch_gim	0.417	0.396	0.750	0.701	0.271	0.375	0.167
zip_pouch_tianji	0.582	0.500	0.873	0.765	0.307	0.255	0.346
Pooled (valid)	0.608	0.453	0.508	0.715	0.315	0.277	0.335
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 14: Complete scenario-level evaluation details for Qwen3-VL-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.756	0.488	0.524	0.455	0.285	0.341	0.407
box_pen_hand	0.642	0.350	0.225	0.275	–	0.250	0.050
box_pen_tianjihand	0.453	0.160	0.060	0.393	0.320	0.250	0.100
box_shoe_tianji	0.941	0.987	0.924	0.496	0.286	0.436	0.420
box_shoes_gim	0.694	0.612	0.684	0.320	0.299	0.327	0.395
brew_teabag_gim	0.598	0.402	0.356	0.288	0.242	0.242	0.167
brew_teabag_tianji	0.596	0.481	0.417	0.744	0.288	0.276	0.359
cap_pen_gim	0.278	0.139	0.244	0.489	0.233	0.233	0.106
make_tea_gim	0.314	0.096	0.173	0.500	0.282	0.282	0.071
make_tea_tianji	0.786	0.256	0.291	0.615	0.368	0.231	0.513
move_flower_hand	0.763	–	–	–	–	–	0.213
move_flower_tianjihand	0.722	0.433	0.233	0.289	0.289	0.400	0.311
move_gift_tianjihand	0.344	0.500	0.583	0.400	0.356	0.233	0.111
pack_aidkit_gim	0.300	0.193	0.200	0.440	0.340	0.327	0.060
pack_aidkit_tianji	0.525	0.250	0.267	0.508	0.350	0.317	0.350
pack_airpods_gim	0.426	0.204	0.333	0.481	0.204	0.315	0.333
pack_airpods_tianji	0.242	0.125	0.275	0.350	0.192	0.250	0.008
pack_express_tianji	0.380	0.480	0.547	0.813	0.293	0.350	0.247
pack_gift_gim	0.280	0.400	0.347	0.260	0.247	0.347	0.080
pack_gift_hand	0.125	0.233	0.400	0.358	–	0.333	0.050
pack_gift_tianji	0.569	0.268	0.325	0.407	0.236	0.325	0.081
pack_sunglasses_hand	0.659	0.220	0.106	0.163	–	0.317	0.358
slip_tshirt_gim	0.313	0.542	0.535	0.354	0.340	0.292	0.181
sort_cubes_gim	0.429	0.548	0.599	0.119	0.243	0.237	0.169
sort_cubes_tianji	0.496	0.252	0.171	0.317	0.407	0.228	0.033
stack_cubes_gim	0.660	0.247	0.240	0.360	0.300	0.260	0.253
stack_cubes_hand	0.729	0.318	0.271	0.240	–	0.233	0.202
stack_cubes_tianji	0.813	0.267	0.253	0.327	0.300	0.210	0.513
stack_cubes_tianjihand	0.880	0.367	0.413	0.320	0.320	0.210	0.300
stow_sunglasses_tianjihand	0.327	0.480	0.510	0.387	0.313	0.300	0.087
takeout_trash_tianji	0.915	0.658	0.564	0.479	0.350	0.231	0.120
tidy_stationery_gim	0.327	0.116	0.278	0.358	0.275	0.222	0.312
tidy_stationery_tianji	0.169	0.444	0.349	0.397	0.265	0.349	0.206
wash_dishes_gim	0.537	0.197	0.190	0.259	0.238	0.184	0.327
wash_dishes_tianji	0.550	0.317	0.342	0.367	0.292	0.167	0.308
wipe_plate_gim	0.393	0.273	0.173	0.393	0.300	0.233	0.173
wipe_plate_tianji	0.708	0.183	0.233	0.383	0.325	0.258	0.275
zip_pouch_gim	0.549	0.188	0.406	0.444	0.236	0.313	0.194
zip_pouch_tianji	0.425	0.118	0.225	0.556	0.268	0.255	0.275
Pooled (valid)	0.509	0.323	0.335	0.396	0.289	0.270	0.223
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 15: Complete scenario-level evaluation details for Qwen3-VL-235B-A22B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.903	0.500	0.650	0.806	0.258	0.600	0.710
box_pen_hand	0.900	0.600	0.667	0.833	–	0.450	0.500
box_pen_tianjihand	0.421	0.316	0.263	1.000	0.579	0.400	0.395
box_shoe_tianji	1.000	1.000	1.000	0.621	0.500	0.600	0.400
box_shoes_gim	1.000	1.000	1.000	0.730	0.432	0.417	0.459
brew_teabag_gim	0.545	0.424	0.455	0.818	0.424	0.273	0.242
brew_teabag_tianji	0.744	0.385	0.487	0.949	0.231	0.333	0.718
cap_pen_gim	0.689	0.333	0.600	0.956	0.311	0.333	0.400
make_tea_gim	0.385	0.385	0.231	0.795	0.359	0.179	0.436
make_tea_tianji	1.000	0.241	0.414	0.931	0.310	0.276	0.897
move_flower_hand	1.000	–	–	–	–	–	0.050
move_flower_tianjihand	0.682	0.667	1.000	0.409	0.409	0.250	0.500
move_gift_tianjihand	0.227	0.800	0.600	0.636	0.455	0.375	0.182
pack_aidkit_gim	0.816	0.447	0.526	0.658	0.658	0.263	0.421
pack_aidkit_tianji	0.667	0.533	0.667	0.800	0.433	0.433	0.200
pack_airpods_gim	0.714	0.429	0.500	0.857	0.429	0.429	0.429
pack_airpods_tianji	0.433	0.200	0.300	0.967	0.233	0.267	0.100
pack_express_tianji	0.447	0.211	0.289	0.868	0.263	0.240	0.368
pack_gift_gim	0.342	0.316	0.474	0.763	0.368	0.316	0.211
pack_gift_hand	0.300	0.267	0.400	0.933	–	0.367	0.400
pack_gift_tianji	0.516	0.258	0.387	0.774	0.452	0.419	0.387
pack_sunglasses_hand	0.871	0.645	0.839	0.774	–	0.258	0.548
slip_tshirt_gim	0.784	0.514	0.784	0.568	0.405	0.333	0.306
sort_cubes_gim	0.705	0.795	0.841	0.636	0.455	0.295	0.432
sort_cubes_tianji	0.903	0.774	0.806	0.968	0.387	0.290	0.290
stack_cubes_gim	0.789	0.658	0.816	0.737	0.211	0.237	0.500
stack_cubes_hand	0.938	0.750	0.750	0.781	–	0.182	0.375
stack_cubes_tianji	0.947	0.789	0.763	0.684	0.632	0.280	0.658
stack_cubes_tianjihand	0.895	0.632	0.842	1.000	0.684	0.320	0.737
stow_sunglasses_tianjihand	0.500	0.320	0.280	0.947	0.500	0.333	0.526
takeout_trash_tianji	1.000	0.897	0.900	0.933	0.393	0.233	0.207
tidy_stationery_gim	0.642	0.407	0.574	0.716	0.395	0.333	0.704
tidy_stationery_tianji	0.596	0.298	0.638	1.000	0.213	0.406	0.638
wash_dishes_gim	0.649	0.243	0.216	0.568	0.405	0.216	0.297
wash_dishes_tianji	0.700	0.433	0.433	0.667	0.533	0.333	0.133
wipe_plate_gim	0.763	0.237	0.132	0.658	0.158	0.211	0.421
wipe_plate_tianji	0.700	0.300	0.333	0.533	0.433	0.167	0.400
zip_pouch_gim	0.528	0.208	0.417	0.889	0.278	0.417	0.500
zip_pouch_tianji	0.763	0.308	0.462	0.974	0.237	0.154	0.500
Pooled (valid)	0.697	0.471	0.561	0.797	0.392	0.304	0.445
Valid record count	1358	1225	1226	1337	1213	1000	1356

Table 16: Complete scenario-level evaluation details for MiMo-VL-7B-RL.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.870	0.537	0.561	0.821	0.301	0.293	0.114
box_pen_hand	0.750	0.283	0.408	0.742	–	0.313	0.050
box_pen_tianjihand	0.513	0.193	0.320	0.947	0.220	0.300	0.033
box_shoe_tianji	0.983	0.975	1.000	0.513	0.319	0.359	0.042
box_shoes_gim	0.850	0.755	0.806	0.680	0.272	0.388	0.204
brew_teabag_gim	0.470	0.235	0.326	0.780	0.220	0.265	0.076
brew_teabag_tianji	0.731	0.327	0.385	0.994	0.365	0.308	0.013
cap_pen_gim	0.278	0.217	0.306	0.978	0.228	0.267	0.050
make_tea_gim	0.212	0.269	0.231	0.949	0.288	0.250	0.051
make_tea_tianji	0.932	0.487	0.265	0.872	0.376	0.316	0.000
move_flower_hand	0.850	–	–	–	–	–	0.000
move_flower_tianjihand	0.622	0.550	0.617	0.478	0.378	0.300	0.200
move_gift_tianjihand	0.533	0.633	0.567	0.611	0.289	0.267	0.300
pack_aidkit_gim	0.673	0.427	0.473	0.747	0.380	0.333	0.340
pack_aidkit_tianji	0.592	0.533	0.558	0.700	0.450	0.283	0.117
pack_airpods_gim	0.556	0.130	0.296	0.741	0.204	0.259	0.204
pack_airpods_tianji	0.383	0.183	0.142	1.000	0.250	0.300	0.008
pack_express_tianji	0.300	0.460	0.507	0.907	0.280	0.300	0.073
pack_gift_gim	0.220	0.427	0.413	0.640	0.207	0.280	0.093
pack_gift_hand	0.292	0.275	0.333	0.825	–	0.300	0.025
pack_gift_tianji	0.455	0.358	0.382	0.715	0.244	0.415	0.000
pack_sunglasses_hand	0.740	0.431	0.520	0.593	–	0.317	0.163
slip_tshirt_gim	0.563	0.438	0.444	0.826	0.271	0.292	0.271
sort_cubes_gim	0.407	0.520	0.531	0.678	0.328	0.271	0.153
sort_cubes_tianji	0.724	0.805	0.748	0.976	0.301	0.317	0.033
stack_cubes_gim	0.507	0.500	0.533	0.813	0.347	0.327	0.260
stack_cubes_hand	0.698	0.705	0.752	0.667	–	0.372	0.000
stack_cubes_tianji	0.973	0.653	0.600	0.613	0.267	0.320	0.000
stack_cubes_tianjihand	0.773	0.660	0.733	0.933	0.393	0.330	0.013
stow_sunglasses_tianjihand	0.367	0.560	0.750	0.707	0.320	0.320	0.187
takeout_trash_tianji	0.846	0.650	0.658	0.872	0.350	0.239	0.000
tidy_stationery_gim	0.528	0.347	0.449	0.707	0.222	0.241	0.290
tidy_stationery_tianji	0.307	0.418	0.429	1.000	0.212	0.317	0.016
wash_dishes_gim	0.646	0.293	0.279	0.837	0.231	0.279	0.027
wash_dishes_tianji	0.675	0.292	0.325	0.617	0.350	0.208	0.000
wipe_plate_gim	0.333	0.253	0.213	0.693	0.253	0.267	0.087
wipe_plate_tianji	0.733	0.233	0.258	0.567	0.233	0.217	0.000
zip_pouch_gim	0.410	0.406	0.573	0.736	0.181	0.333	0.090
zip_pouch_tianji	0.601	0.343	0.529	0.804	0.216	0.235	0.209
Pooled (valid)	0.570	0.428	0.464	0.778	0.283	0.294	0.103
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 17: Complete scenario-level evaluation details for Cosmos-Reason2-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.732	0.476	0.793	0.293	0.211	0.293	0.813
box_pen_hand	0.783	0.375	0.250	0.617	–	0.238	0.358
box_pen_tianjihand	0.400	0.047	0.087	0.707	0.287	0.200	0.447
box_shoe_tianji	0.882	0.962	0.873	0.650	0.336	0.282	0.664
box_shoes_gim	0.973	0.735	0.582	0.524	0.286	0.408	0.755
brew_teabag_gim	0.371	0.477	0.394	0.288	0.258	0.265	0.311
brew_teabag_tianji	0.660	0.513	0.513	0.699	0.282	0.282	0.462
cap_pen_gim	0.411	0.128	0.350	0.372	0.272	0.258	0.450
make_tea_gim	0.346	0.135	0.237	0.410	0.199	0.276	0.135
make_tea_tianji	0.735	0.462	0.462	0.658	0.393	0.171	0.778
move_flower_hand	0.988	–	–	–	–	–	0.550
move_flower_tianjihand	0.811	0.700	0.650	0.278	0.267	0.267	0.356
move_gift_tianjihand	0.478	0.683	0.617	0.433	0.389	0.367	0.522
pack_aidkit_gim	0.633	0.360	0.393	0.533	0.353	0.247	0.193
pack_aidkit_tianji	0.567	0.350	0.275	0.675	0.375	0.275	0.375
pack_airpods_gim	0.352	0.278	0.370	0.352	0.278	0.315	0.537
pack_airpods_tianji	0.342	0.183	0.292	0.650	0.225	0.267	0.008
pack_express_tianji	0.387	0.393	0.627	0.733	0.267	0.340	0.240
pack_gift_gim	0.347	0.473	0.360	0.320	0.160	0.313	0.080
pack_gift_hand	0.217	0.283	0.342	0.633	–	0.217	0.075
pack_gift_tianji	0.488	0.325	0.268	0.496	0.496	0.252	0.260
pack_sunglasses_hand	0.724	0.512	0.211	0.317	–	0.285	0.407
slip_tshirt_gim	0.285	0.507	0.611	0.250	0.306	0.281	0.465
sort_cubes_gim	0.469	0.638	0.638	0.271	0.333	0.215	0.390
sort_cubes_tianji	0.626	0.626	0.569	0.398	0.276	0.268	0.033
stack_cubes_gim	0.820	0.553	0.520	0.313	0.307	0.293	0.660
stack_cubes_hand	0.868	0.504	0.674	0.465	–	0.244	0.403
stack_cubes_tianji	0.980	0.687	0.640	0.380	0.380	0.240	0.493
stack_cubes_tianjihand	0.973	0.653	0.707	0.607	0.353	0.300	0.440
stow_sunglasses_tianjihand	0.373	0.620	0.410	0.660	0.260	0.240	0.373
takeout_trash_tianji	0.906	0.761	0.709	0.906	0.359	0.162	0.154
tidy_stationery_gim	0.417	0.389	0.472	0.173	0.306	0.343	0.478
tidy_stationery_tianji	0.360	0.603	0.534	0.471	0.238	0.405	0.217
wash_dishes_gim	0.660	0.259	0.272	0.190	0.347	0.211	0.361
wash_dishes_tianji	0.633	0.325	0.325	0.558	0.400	0.275	0.325
wipe_plate_gim	0.367	0.247	0.140	0.347	0.293	0.293	0.340
wipe_plate_tianji	0.725	0.242	0.192	0.325	0.425	0.350	0.250
zip_pouch_gim	0.611	0.271	0.740	0.438	0.208	0.250	0.278
zip_pouch_tianji	0.431	0.284	0.814	0.725	0.333	0.294	0.556
Pooled (valid)	0.578	0.435	0.458	0.465	0.304	0.271	0.383
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 18: Complete scenario-level evaluation details for Cosmos3-Edge-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.569	0.268	0.317	0.374	0.236	0.171	0.374
box_pen_hand	0.933	0.408	0.425	0.300	–	0.250	0.467
box_pen_tianjihand	0.333	0.293	0.287	0.173	0.193	0.190	–
box_shoe_tianji	0.714	0.506	0.456	0.547	0.218	0.282	0.605
box_shoes_gim	0.633	0.694	0.694	0.327	0.224	0.306	0.653
brew_teabag_gim	0.341	0.326	0.333	0.235	0.212	0.205	0.038
brew_teabag_tianji	0.609	0.436	0.429	0.474	0.224	–	0.308
cap_pen_gim	0.339	0.294	0.306	0.094	0.194	0.258	0.144
make_tea_gim	0.154	0.256	0.353	0.218	0.160	0.186	0.038
make_tea_tianji	0.675	0.333	0.368	0.615	0.291	0.265	0.350
move_flower_hand	1.000	–	–	–	–	–	0.400
move_flower_tianjihand	0.422	0.350	0.283	0.167	0.211	0.133	0.178
move_gift_tianjihand	0.422	0.633	0.667	0.311	0.222	0.100	0.067
pack_aidkit_gim	0.513	0.380	0.307	0.193	0.220	0.260	0.087
pack_aidkit_tianji	0.633	0.475	0.433	0.258	0.292	0.217	0.275
pack_airpods_gim	0.315	0.185	0.204	0.241	0.111	0.148	0.130
pack_airpods_tianji	0.408	0.158	0.242	0.058	0.300	–	0.092
pack_express_tianji	0.260	0.307	0.347	0.240	0.127	0.280	0.140
pack_gift_gim	0.240	0.460	0.353	0.073	0.253	0.227	0.167
pack_gift_hand	0.367	0.300	0.375	0.200	–	0.242	0.175
pack_gift_tianji	0.285	0.187	0.236	0.431	0.325	0.163	0.089
pack_sunglasses_hand	0.935	0.561	0.455	0.228	–	0.220	0.350
slip_tshirt_gim	0.486	0.500	0.618	0.208	0.222	0.177	0.417
sort_cubes_gim	0.429	0.588	0.588	0.068	0.186	0.175	0.249
sort_cubes_tianji	0.634	0.650	0.520	0.203	0.211	–	0.138
stack_cubes_gim	0.580	0.533	0.587	0.120	0.247	0.200	0.453
stack_cubes_hand	0.969	0.612	0.752	0.171	–	0.105	0.140
stack_cubes_tianji	0.660	0.820	0.553	0.100	0.227	0.190	0.580
stack_cubes_tianjihand	0.660	0.487	0.547	0.273	0.220	–	0.260
stow_sunglasses_tianjihand	0.433	0.380	0.480	0.240	0.253	0.120	0.193
takeout_trash_tianji	0.675	0.410	0.393	0.402	0.282	0.248	0.256
tidy_stationery_gim	0.478	0.343	0.472	0.176	0.179	0.194	0.222
tidy_stationery_tianji	0.381	0.370	0.370	0.243	0.138	0.167	0.048
wash_dishes_gim	0.401	0.231	0.272	0.109	0.252	0.204	0.259
wash_dishes_tianji	0.392	0.308	0.358	0.225	0.275	0.167	0.225
wipe_plate_gim	0.333	0.220	0.280	0.147	0.260	0.140	0.347
wipe_plate_tianji	0.425	0.150	0.183	0.350	0.217	0.183	0.142
zip_pouch_gim	0.472	0.479	0.510	0.257	0.181	0.208	0.368
zip_pouch_tianji	0.542	0.353	0.647	0.288	0.261	0.294	0.216
Pooled (valid)	0.503	0.401	0.421	0.237	0.222	0.204	0.253
Valid record count	5407	4880	4880	5325	4835	3480	5257

Table 19: Complete scenario-level evaluation details for SenseNova-SI-1.1-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.236	0.439	0.512	0.333	0.260	0.293	0.000
box_pen_hand	0.458	0.208	0.367	0.533	–	0.275	0.000
box_pen_tianjihand	0.500	0.213	0.347	0.473	0.273	0.310	0.000
box_shoe_tianji	0.899	0.696	0.975	0.368	0.277	0.436	0.000
box_shoes_gim	0.524	0.724	0.857	0.463	0.238	0.367	0.000
brew_teabag_gim	0.295	0.530	0.379	0.394	0.220	0.326	0.000
brew_teabag_tianji	0.500	0.558	0.474	0.487	0.295	0.314	0.000
cap_pen_gim	0.322	0.261	0.350	0.311	0.256	0.325	0.000
make_tea_gim	0.199	0.154	0.147	0.378	0.269	0.237	0.006
make_tea_tianji	0.325	0.410	0.427	0.368	0.333	0.376	0.000
move_flower_hand	0.613	–	–	–	–	–	0.000
move_flower_tianjihand	0.244	0.083	0.500	0.322	0.211	0.433	0.000
move_gift_tianjihand	0.289	0.267	0.733	0.378	0.344	0.300	0.000
pack_aidkit_gim	0.453	0.380	0.407	0.313	0.287	0.267	0.000
pack_aidkit_tianji	0.625	0.283	0.342	0.275	0.308	0.342	0.000
pack_airpods_gim	0.315	0.167	0.315	0.426	0.315	0.315	0.000
pack_airpods_tianji	0.317	0.142	0.292	0.417	0.350	0.317	0.000
pack_express_tianji	0.307	0.300	0.400	0.407	0.220	0.270	0.000
pack_gift_gim	0.313	0.293	0.347	0.340	0.253	0.287	0.000
pack_gift_hand	0.458	0.292	0.350	0.267	–	0.233	0.000
pack_gift_tianji	0.285	0.203	0.333	0.317	0.211	0.358	0.000
pack_sunglasses_hand	0.431	0.244	0.171	0.431	–	0.309	0.000
slip_tshirt_gim	0.549	0.243	0.465	0.313	0.250	0.281	0.035
sort_cubes_gim	0.418	0.548	0.486	0.367	0.311	0.220	0.000
sort_cubes_tianji	0.382	0.642	0.463	0.341	0.350	0.260	0.000
stack_cubes_gim	0.207	0.260	0.207	0.513	0.267	0.267	0.000
stack_cubes_hand	0.481	0.450	0.372	0.512	–	0.302	0.000
stack_cubes_tianji	0.460	0.447	0.407	0.467	0.287	0.360	0.000
stack_cubes_tianjihand	0.567	0.560	0.607	0.387	0.320	0.240	0.000
stow_sunglasses_tianjihand	0.440	0.540	0.560	0.413	0.200	0.320	0.000
takeout_trash_tianji	0.615	0.573	0.470	0.538	0.325	0.162	0.000
tidy_stationery_gim	0.410	0.278	0.755	0.352	0.231	0.222	0.000
tidy_stationery_tianji	0.280	0.450	0.550	0.439	0.233	0.437	0.000
wash_dishes_gim	0.381	0.109	0.163	0.585	0.204	0.293	0.000
wash_dishes_tianji	0.450	0.267	0.200	0.517	0.267	0.208	0.000
wipe_plate_gim	0.487	0.187	0.113	0.380	0.233	0.180	0.000
wipe_plate_tianji	0.283	0.208	0.225	0.433	0.233	0.275	0.008
zip_pouch_gim	0.257	0.302	0.490	0.375	0.292	0.271	0.000
zip_pouch_tianji	0.431	0.373	0.608	0.261	0.248	0.176	0.000
Pooled (valid)	0.409	0.349	0.415	0.398	0.266	0.286	0.001
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 20: Complete scenario-level evaluation details for InternVL3-2B.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.407	0.402	0.537	0.301	0.179	0.171	0.024
box_pen_hand	0.558	0.217	0.300	0.300	–	0.263	0.000
box_pen_tianjihand	0.487	0.247	0.333	0.313	0.300	0.240	0.007
box_shoe_tianji	0.924	0.962	1.000	0.265	0.286	0.333	0.000
box_shoes_gim	0.605	0.551	0.857	0.265	0.265	0.306	0.136
brew_teabag_gim	0.053	0.477	0.379	0.250	0.273	0.288	0.000
brew_teabag_tianji	0.346	0.571	0.372	0.327	0.231	0.333	0.045
cap_pen_gim	0.322	0.206	0.339	0.272	0.233	0.300	0.006
make_tea_gim	0.058	0.263	0.212	0.276	0.231	0.218	0.077
make_tea_tianji	0.368	0.299	0.402	0.231	0.248	0.402	0.000
move_flower_hand	0.800	–	–	–	–	–	0.000
move_flower_tianjihand	0.444	0.267	0.733	0.267	0.244	0.433	0.022
move_gift_tianjihand	0.422	0.483	0.600	0.322	0.200	0.300	0.000
pack_aidkit_gim	0.387	0.273	0.333	0.253	0.287	0.233	0.080
pack_aidkit_tianji	0.625	0.308	0.333	0.208	0.225	0.308	0.050
pack_airpods_gim	0.315	0.056	0.333	0.185	0.278	0.204	0.056
pack_airpods_tianji	0.283	0.142	0.308	0.275	0.242	0.217	0.000
pack_express_tianji	0.400	0.313	0.387	0.287	0.360	0.310	0.053
pack_gift_gim	0.273	0.207	0.333	0.213	0.273	0.227	0.027
pack_gift_hand	0.300	0.175	0.350	0.275	–	0.308	0.083
pack_gift_tianji	0.398	0.228	0.341	0.244	0.228	0.325	0.024
pack_sunglasses_hand	0.780	0.309	0.504	0.220	–	0.325	0.008
slip_tshirt_gim	0.465	0.472	0.521	0.278	0.257	0.198	0.097
sort_cubes_gim	0.373	0.390	0.373	0.198	0.243	0.220	0.017
sort_cubes_tianji	0.415	0.602	0.520	0.244	0.260	0.252	0.008
stack_cubes_gim	0.380	0.207	0.247	0.307	0.273	0.240	0.000
stack_cubes_hand	0.643	0.318	0.512	0.279	–	0.267	0.000
stack_cubes_tianji	0.620	0.527	0.607	0.247	0.247	0.250	0.020
stack_cubes_tianjihand	0.673	0.687	0.713	0.300	0.227	0.250	0.000
stow_sunglasses_tianjihand	0.227	0.320	0.650	0.260	0.307	0.360	0.000
takeout_trash_tianji	0.658	0.581	0.479	0.282	0.205	0.222	0.009
tidy_stationery_gim	0.438	0.106	0.514	0.244	0.269	0.204	0.022
tidy_stationery_tianji	0.307	0.392	0.439	0.286	0.259	0.341	0.000
wash_dishes_gim	0.408	0.095	0.088	0.190	0.306	0.252	0.000
wash_dishes_tianji	0.492	0.258	0.283	0.258	0.258	0.192	0.000
wipe_plate_gim	0.480	0.273	0.100	0.220	0.260	0.173	0.000
wipe_plate_tianji	0.458	0.300	0.233	0.258	0.267	0.225	0.000
zip_pouch_gim	0.264	0.146	0.542	0.278	0.188	0.292	0.007
zip_pouch_tianji	0.405	0.588	0.804	0.216	0.307	0.196	0.000
Pooled (valid)	0.433	0.340	0.423	0.260	0.258	0.262	0.023
Valid record count	5407	4880	4880	5325	4835	3979	5407

Table 21: Complete scenario-level evaluation details for Doubao-Seed-2.0-Lite.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.903	0.400	0.500	0.903	0.323	0.500	0.742
box_pen_hand	1.000	0.500	0.433	1.000	–	0.400	0.967
box_pen_tianjihand	0.711	0.316	0.579	1.000	0.474	0.400	0.500
box_shoe_tianji	1.000	1.000	1.000	0.759	0.500	0.600	0.367
box_shoes_gim	1.000	0.958	1.000	1.000	0.459	0.250	0.568
brew_teabag_gim	0.424	0.455	0.303	0.939	0.394	0.303	0.303
brew_teabag_tianji	0.718	0.462	0.513	1.000	0.410	0.385	0.744
cap_pen_gim	0.578	0.356	0.600	0.978	0.422	0.333	0.667
make_tea_gim	0.692	0.436	0.436	0.923	0.410	0.128	0.590
make_tea_tianji	1.000	0.276	0.276	0.828	0.345	0.517	0.897
move_flower_hand	1.000	–	–	–	–	–	1.000
move_flower_tianjihand	0.773	0.933	1.000	0.909	0.636	0.500	0.500
move_gift_tianjihand	0.545	0.733	0.867	0.955	0.682	0.375	0.364
pack_aidkit_gim	0.763	0.447	0.658	0.895	0.684	0.316	0.184
pack_aidkit_tianji	0.833	0.667	0.667	0.900	0.367	0.300	0.433
pack_airpods_gim	0.857	0.286	0.500	1.000	0.571	0.357	0.786
pack_airpods_tianji	0.533	0.167	0.400	1.000	0.367	0.300	0.133
pack_express_tianji	0.579	0.316	0.553	0.974	0.184	0.200	0.632
pack_gift_gim	0.553	0.500	0.605	0.868	0.789	0.395	0.474
pack_gift_hand	0.767	0.400	0.533	0.900	–	0.367	0.667
pack_gift_tianji	0.710	0.258	0.419	0.935	0.516	0.484	0.387
pack_sunglasses_hand	0.806	0.645	0.806	0.903	–	0.387	0.871
slip_tshirt_gim	0.622	0.568	0.676	0.811	0.162	0.458	0.472
sort_cubes_gim	0.841	1.000	1.000	0.773	0.545	0.364	0.523
sort_cubes_tianji	0.903	0.968	1.000	1.000	0.290	0.355	0.290
stack_cubes_gim	0.553	0.684	0.842	0.974	0.184	0.289	0.684
stack_cubes_hand	1.000	0.719	1.000	0.906	–	0.273	0.563
stack_cubes_tianji	1.000	0.868	0.789	0.842	0.500	0.440	0.789
stack_cubes_tianjihand	1.000	0.737	0.895	1.000	0.789	0.520	0.763
stow_sunglasses_tianjihand	0.658	0.600	0.680	0.974	0.474	0.417	0.789
takeout_trash_tianji	1.000	0.897	0.900	1.000	0.429	0.133	0.517
tidy_stationery_gim	0.679	0.519	0.648	0.864	0.457	0.519	0.630
tidy_stationery_tianji	0.745	0.213	0.574	1.000	0.511	0.500	0.638
wash_dishes_gim	0.838	0.568	0.459	0.865	0.541	0.243	0.432
wash_dishes_tianji	0.733	0.500	0.467	0.800	0.533	0.433	0.800
wipe_plate_gim	0.763	0.421	0.211	0.842	0.316	0.421	0.447
wipe_plate_tianji	0.433	0.333	0.267	1.000	0.333	0.233	0.200
zip_pouch_gim	0.667	0.250	0.667	0.944	0.278	0.417	0.556
zip_pouch_tianji	0.605	0.500	0.615	0.947	0.553	0.308	0.658
Pooled (valid)	0.754	0.538	0.631	0.921	0.451	0.359	0.577
Valid record count	1358	1225	1226	1337	1213	1000	1356

Table 22: Complete scenario-level evaluation details for Gemini-3.6-Flash.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.355	0.550	0.550	0.161	0.613	0.600	0.419
box_pen_hand	0.300	0.367	0.067	0.100	–	0.500	0.067
box_pen_tianjihand	0.289	0.342	0.026	0.316	0.632	0.680	0.579
box_shoe_tianji	0.400	0.550	0.550	0.310	0.733	0.900	0.133
box_shoes_gim	0.486	0.792	0.542	0.459	0.541	0.667	0.243
brew_teabag_gim	0.091	0.394	0.455	0.333	0.606	0.303	0.212
brew_teabag_tianji	0.103	0.282	0.359	0.359	0.564	0.385	0.000
cap_pen_gim	0.267	0.222	0.444	0.200	0.467	0.500	0.289
make_tea_gim	0.103	0.154	0.128	0.205	0.513	0.205	0.026
make_tea_tianji	0.448	0.172	0.483	0.241	0.862	0.414	0.000
move_flower_hand	0.550	–	–	–	–	–	0.550
move_flower_tianjihand	0.364	0.067	0.000	0.182	0.591	0.750	0.500
move_gift_tianjihand	0.318	0.000	0.600	0.182	0.682	0.750	0.636
pack_aidkit_gim	0.289	0.237	0.184	0.132	0.737	0.553	0.000
pack_aidkit_tianji	0.233	0.033	0.133	0.200	0.333	0.600	0.000
pack_airpods_gim	0.214	0.429	0.286	0.357	0.500	0.357	0.000
pack_airpods_tianji	0.267	0.267	0.433	0.133	0.200	0.267	0.000
pack_express_tianji	0.237	0.237	0.474	0.211	0.474	0.080	0.211
pack_gift_gim	0.053	0.316	0.395	0.211	0.789	0.368	0.000
pack_gift_hand	0.200	0.267	0.367	0.133	–	0.400	0.000
pack_gift_tianji	0.323	0.258	0.387	0.194	0.677	0.613	0.000
pack_sunglasses_hand	0.226	0.065	0.258	0.129	–	0.548	0.000
slip_tshirt_gim	0.054	0.270	0.405	0.351	0.378	0.708	0.528
sort_cubes_gim	0.273	0.114	0.114	0.159	0.705	0.750	0.114
sort_cubes_tianji	0.129	0.000	0.065	0.452	0.839	0.710	0.032
stack_cubes_gim	0.605	0.421	0.316	0.263	0.132	0.553	0.158
stack_cubes_hand	0.344	0.281	0.531	0.250	–	0.773	0.125
stack_cubes_tianji	0.342	0.132	0.079	0.211	0.895	0.880	0.105
stack_cubes_tianjihand	0.474	0.237	0.316	0.211	0.895	0.800	0.395
stow_sunglasses_tianjihand	0.342	0.440	0.480	0.395	0.737	0.417	0.395
takeout_trash_tianji	0.200	0.241	0.200	0.233	0.214	0.600	0.000
tidy_stationery_gim	0.358	0.556	0.685	0.321	0.630	0.481	0.284
tidy_stationery_tianji	0.149	0.511	0.426	0.340	0.617	0.563	0.149
wash_dishes_gim	0.297	0.297	0.351	0.216	0.811	0.270	0.000
wash_dishes_tianji	0.067	0.433	0.367	0.067	0.733	0.333	0.133
wipe_plate_gim	0.184	0.211	0.237	0.237	0.342	0.316	0.342
wipe_plate_tianji	0.300	0.300	0.133	0.100	0.800	0.333	0.133
zip_pouch_gim	0.500	0.708	0.917	0.139	0.583	0.500	0.194
zip_pouch_tianji	0.368	0.731	0.731	0.421	0.763	0.308	0.289
Pooled (valid)	0.284	0.308	0.347	0.245	0.608	0.496	0.187
Valid record count	1358	1225	1226	1337	1213	1000	1356

Table 23: Complete scenario-level evaluation details for GPT-6-Astra.

Scenario	Current Action	Next Action	Next +Goal	Frame Match	View Match	Frame Order	Action Time
bag_toy_gim	0.938	0.778	0.904	0.984	0.669	0.705	0.822
box_pen_hand	0.951	0.742	0.792	0.989	–	0.689	0.847
box_pen_tianjihand	0.852	0.627	0.714	0.991	0.701	0.702	0.770
box_shoe_tianji	0.986	0.977	0.984	0.981	0.732	0.789	0.804
box_shoes_gim	0.974	0.958	0.971	0.982	0.694	0.729	0.835
brew_teabag_gim	0.832	0.765	0.793	0.982	0.644	0.660	0.593
brew_teabag_tianji	0.901	0.790	0.819	0.993	0.653	0.704	0.806
cap_pen_gim	0.837	0.641	0.817	0.987	0.588	0.696	0.739
make_tea_gim	0.763	0.574	0.614	0.987	0.611	0.603	0.637
make_tea_tianji	0.966	0.724	0.786	0.988	0.721	0.695	0.899
move_flower_hand	0.977	–	–	–	–	–	0.873
move_flower_tianjihand	0.912	0.857	0.907	0.965	0.705	0.766	0.739
move_gift_tianjihand	0.796	0.883	0.943	0.980	0.714	0.719	0.611
pack_aidkit_gim	0.912	0.760	0.819	0.979	0.765	0.687	0.590
pack_aidkit_tianji	0.916	0.825	0.870	0.982	0.676	0.726	0.604
pack_airpods_gim	0.861	0.630	0.758	0.985	0.660	0.690	0.708
pack_airpods_tianji	0.847	0.573	0.727	0.989	0.561	0.636	0.292
pack_express_tianji	0.839	0.662	0.807	0.989	0.577	0.647	0.747
pack_gift_gim	0.762	0.785	0.819	0.976	0.687	0.675	0.510
pack_gift_hand	0.802	0.644	0.782	0.986	–	0.703	0.626
pack_gift_tianji	0.873	0.626	0.726	0.983	0.716	0.749	0.658
pack_sunglasses_hand	0.946	0.834	0.882	0.978	–	0.713	0.795
slip_tshirt_gim	0.881	0.795	0.906	0.979	0.607	0.689	0.787
sort_cubes_gim	0.907	0.902	0.927	0.971	0.720	0.663	0.744
sort_cubes_tianji	0.938	0.921	0.933	0.988	0.730	0.688	0.629
stack_cubes_gim	0.913	0.831	0.872	0.983	0.572	0.669	0.825
stack_cubes_hand	0.967	0.861	0.927	0.985	–	0.712	0.815
stack_cubes_tianji	0.977	0.892	0.911	0.982	0.745	0.691	0.888
stack_cubes_tianjihand	0.975	0.861	0.931	0.989	0.796	0.699	0.842
stow_sunglasses_tianjihand	0.835	0.844	0.898	0.988	0.714	0.698	0.794
takeout_trash_tianji	0.967	0.912	0.920	0.990	0.651	0.569	0.670
tidy_stationery_gim	0.879	0.711	0.872	0.978	0.652	0.651	0.789
tidy_stationery_tianji	0.815	0.798	0.871	0.990	0.644	0.755	0.710
wash_dishes_gim	0.919	0.646	0.722	0.976	0.726	0.615	0.685
wash_dishes_tianji	0.916	0.746	0.798	0.979	0.732	0.614	0.714
wipe_plate_gim	0.875	0.638	0.599	0.973	0.600	0.610	0.662
wipe_plate_tianji	0.899	0.589	0.662	0.976	0.710	0.607	0.542
zip_pouch_gim	0.868	0.688	0.886	0.983	0.601	0.728	0.737
zip_pouch_tianji	0.892	0.742	0.899	0.988	0.676	0.613	0.781
Scenario macro-mean	0.894	0.764	0.836	0.983	0.675	0.683	0.721